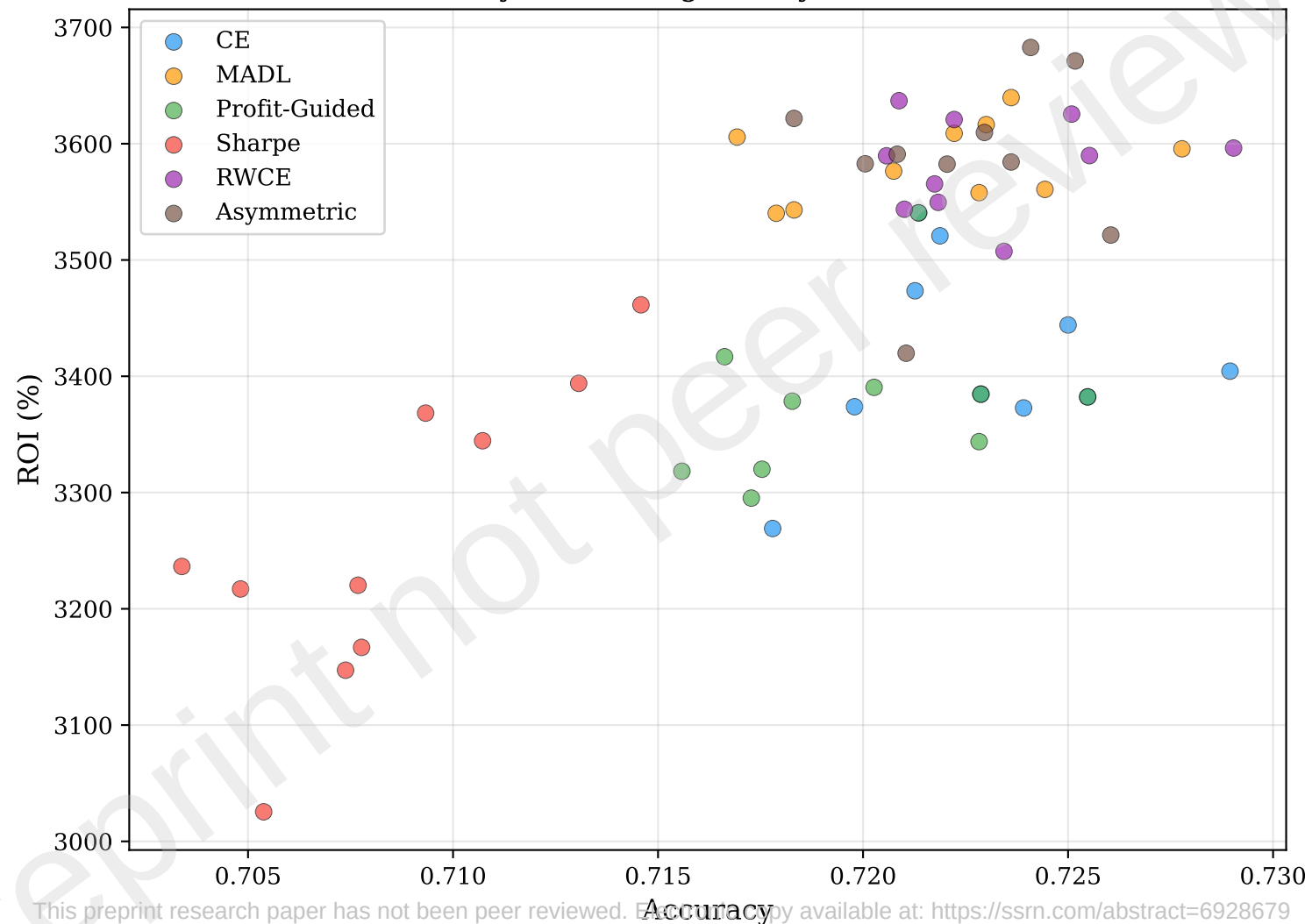
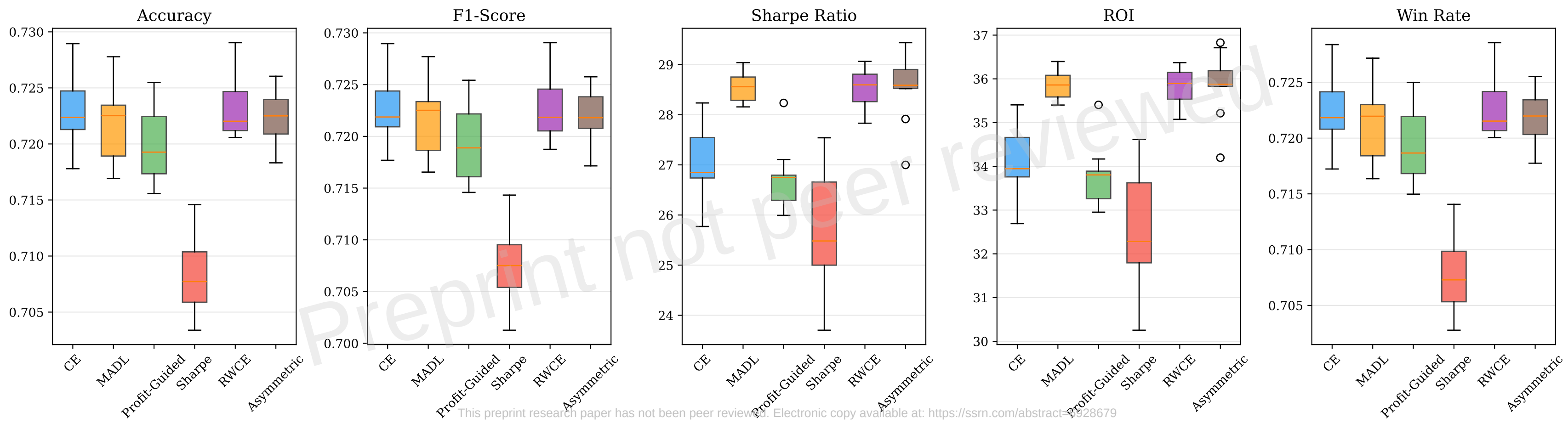


Accuracy vs. Trading ROI by Loss Function

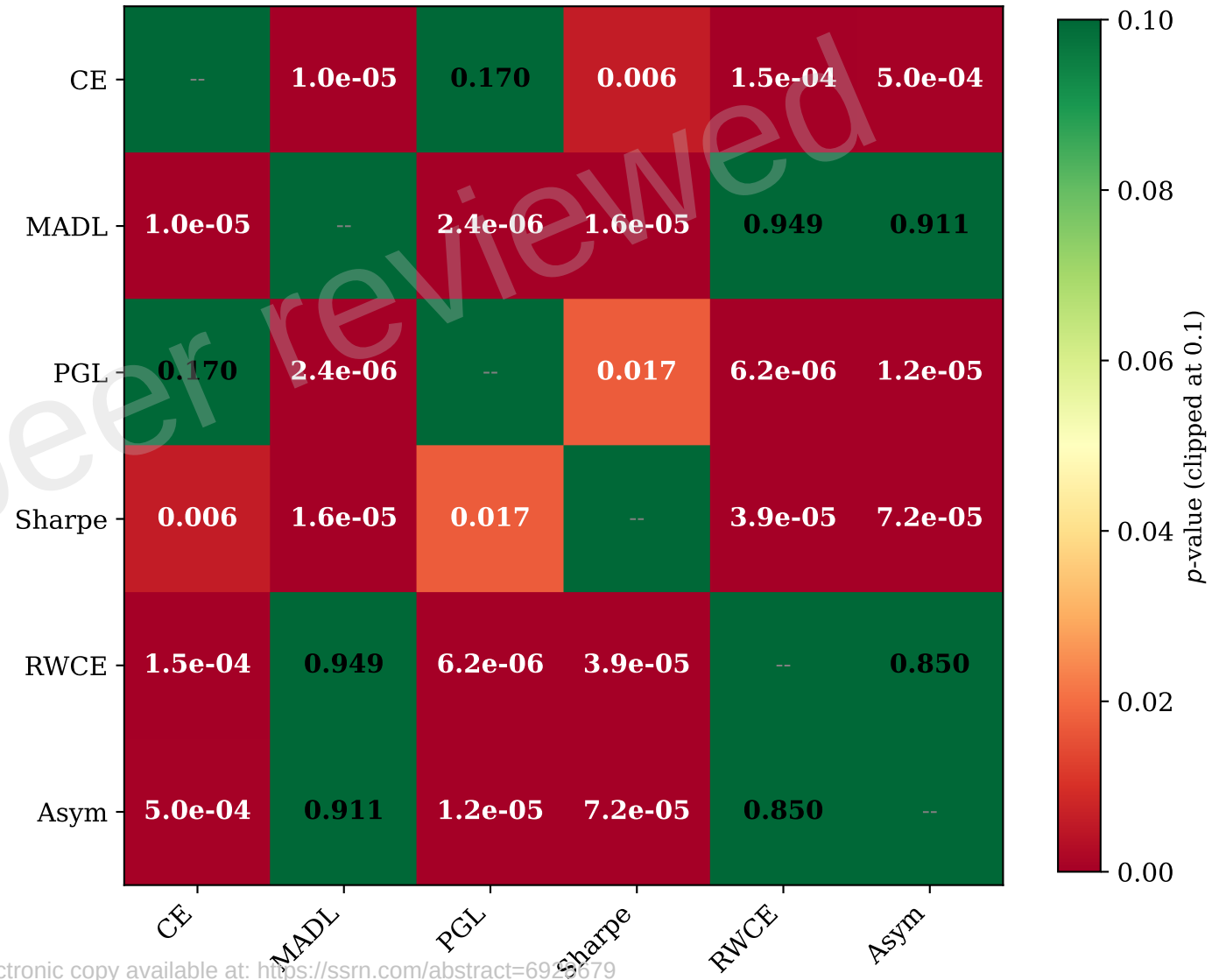




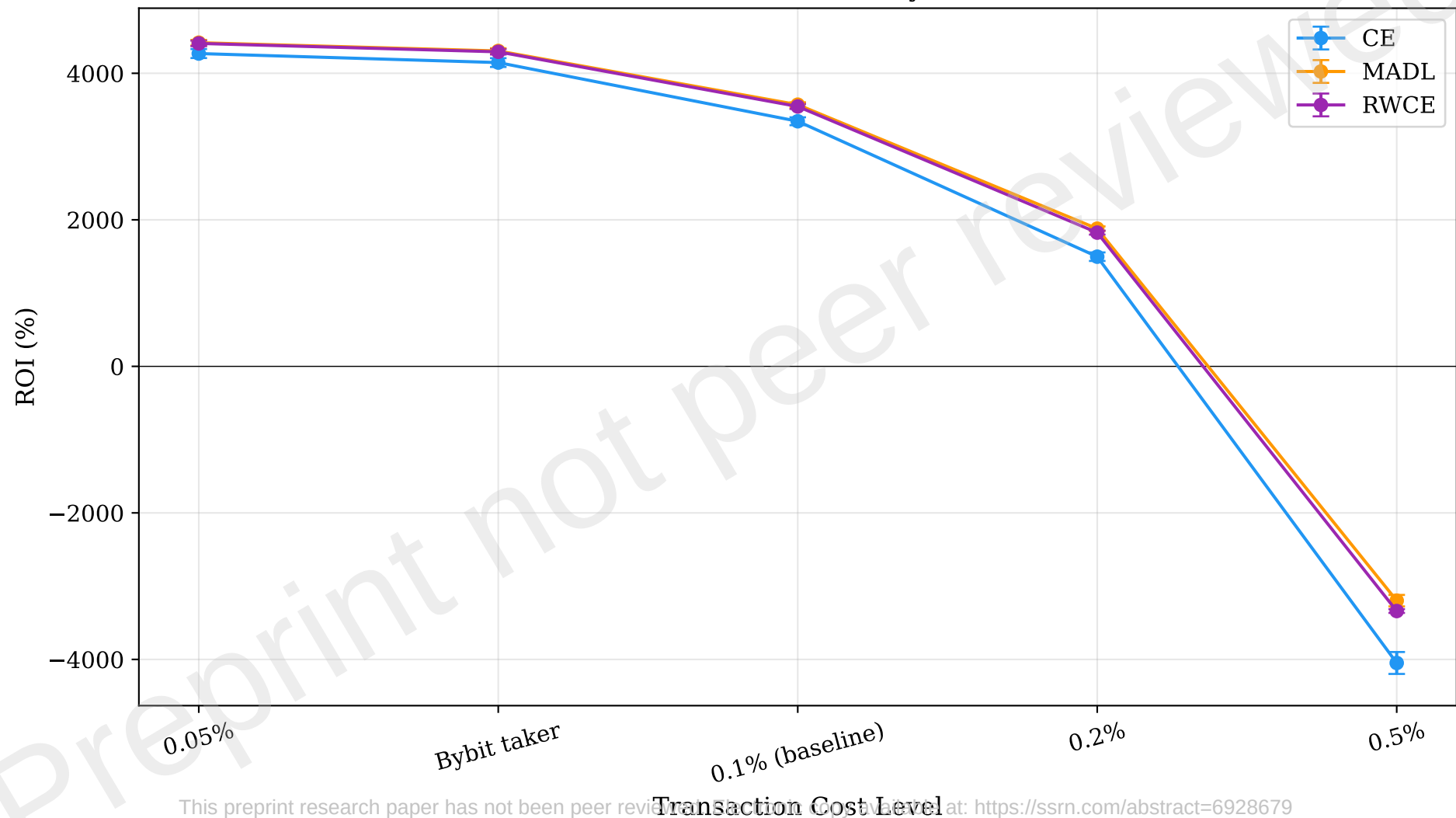
Paired *t*-test *p*-values: Accuracy



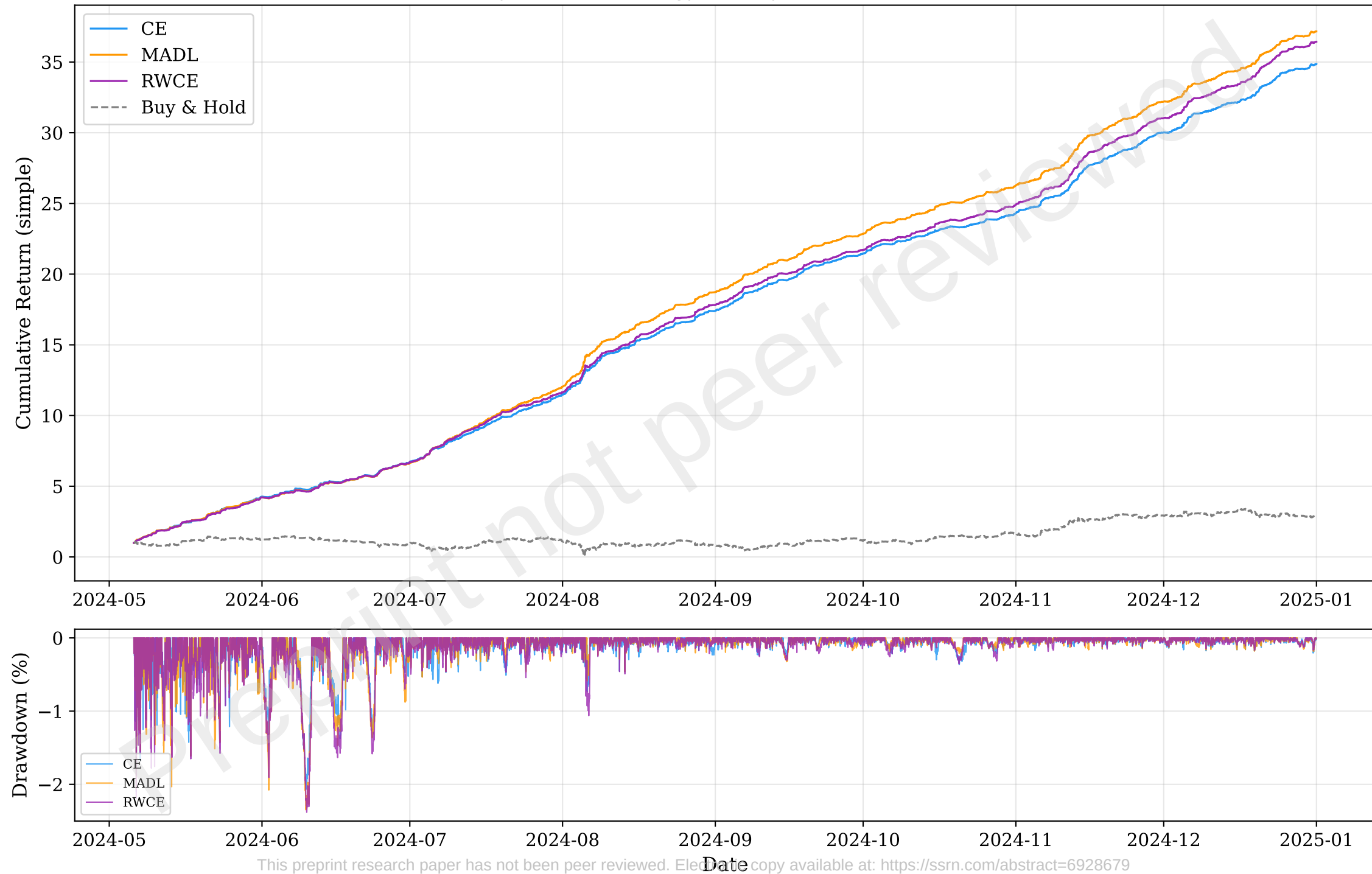
Paired *t*-test *p*-values: Sharpe ratio



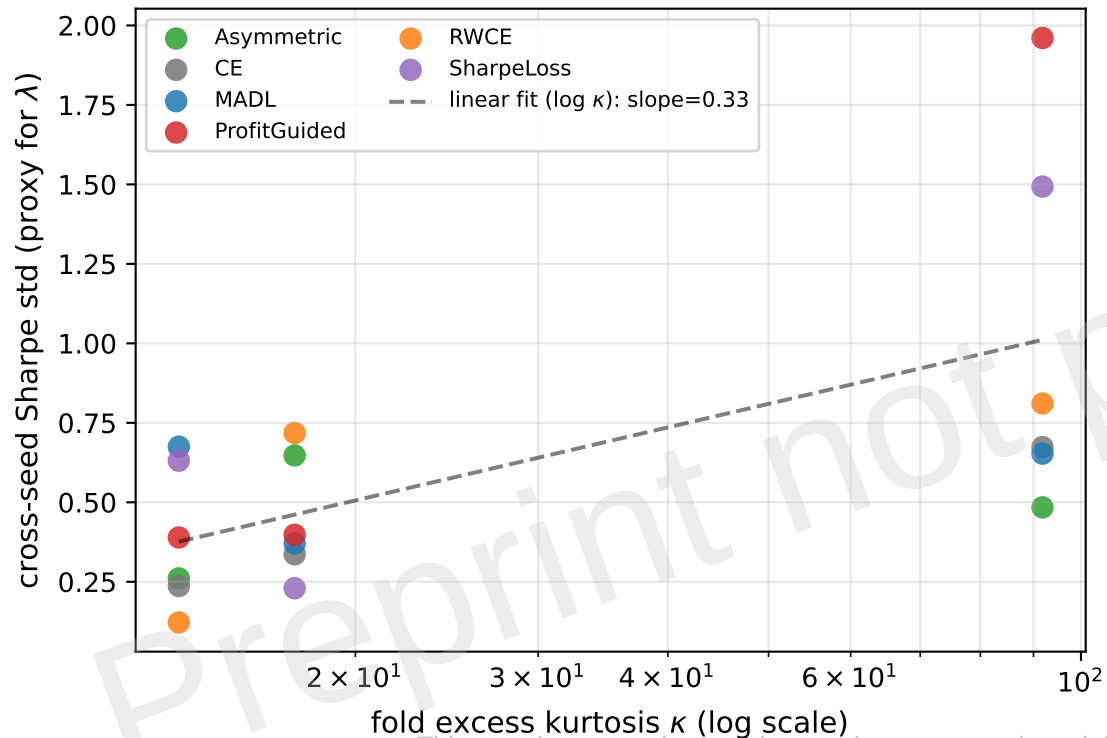
Transaction Cost Sensitivity (3 seeds)



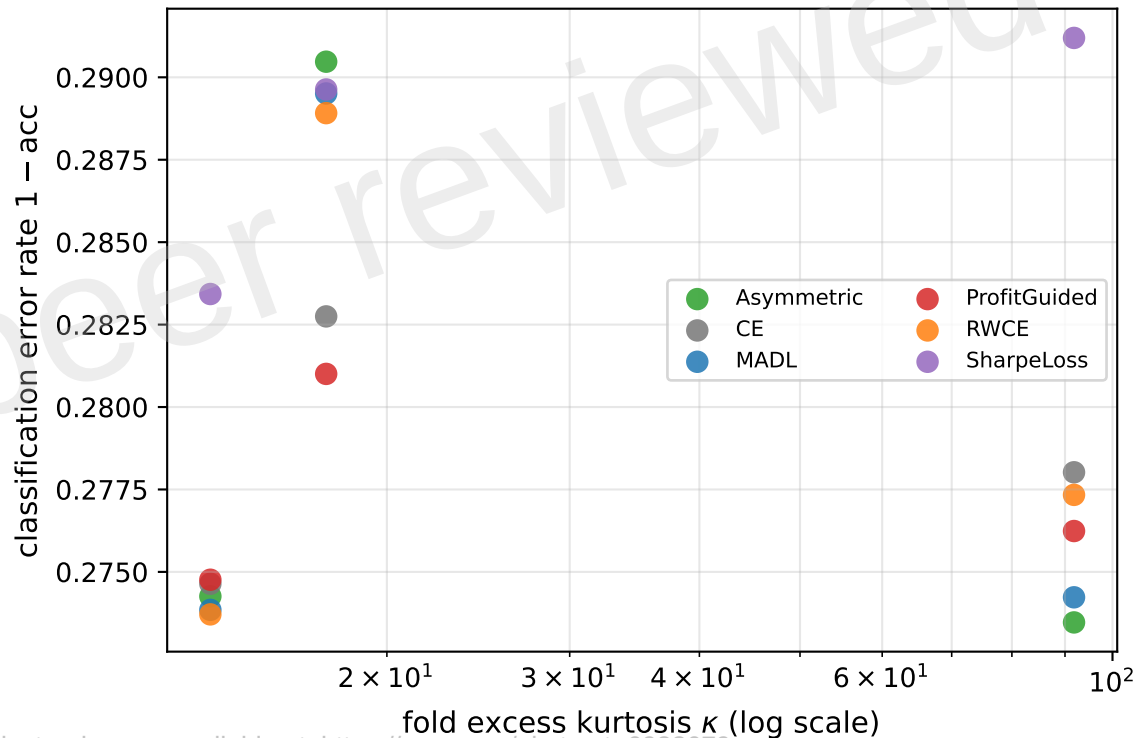
Equity Curves: Strategy vs. Buy & Hold (Test Period)



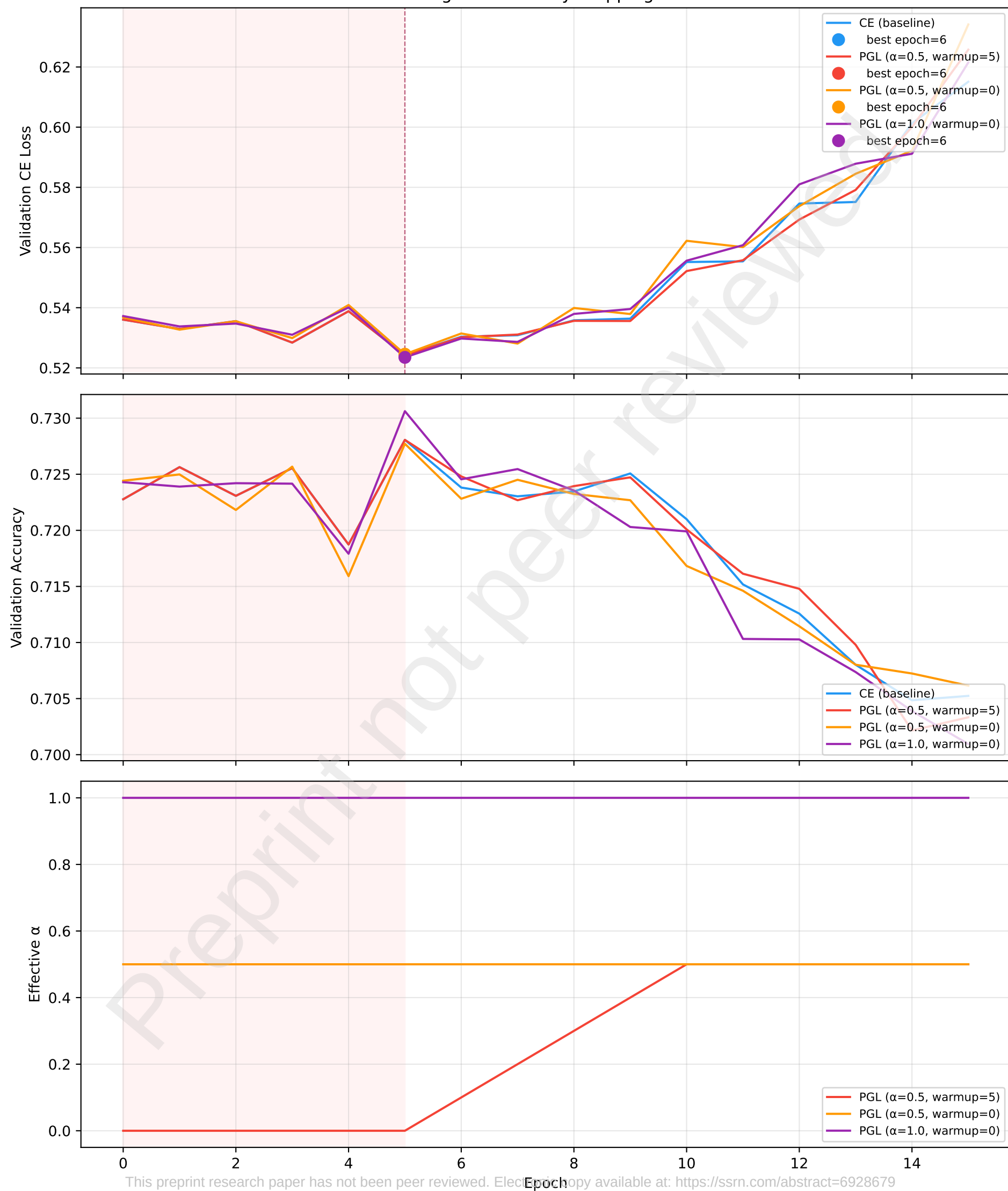
(a) λ proxy vs κ (n=18, Pearson $r=+0.633$, $p=0.005$)



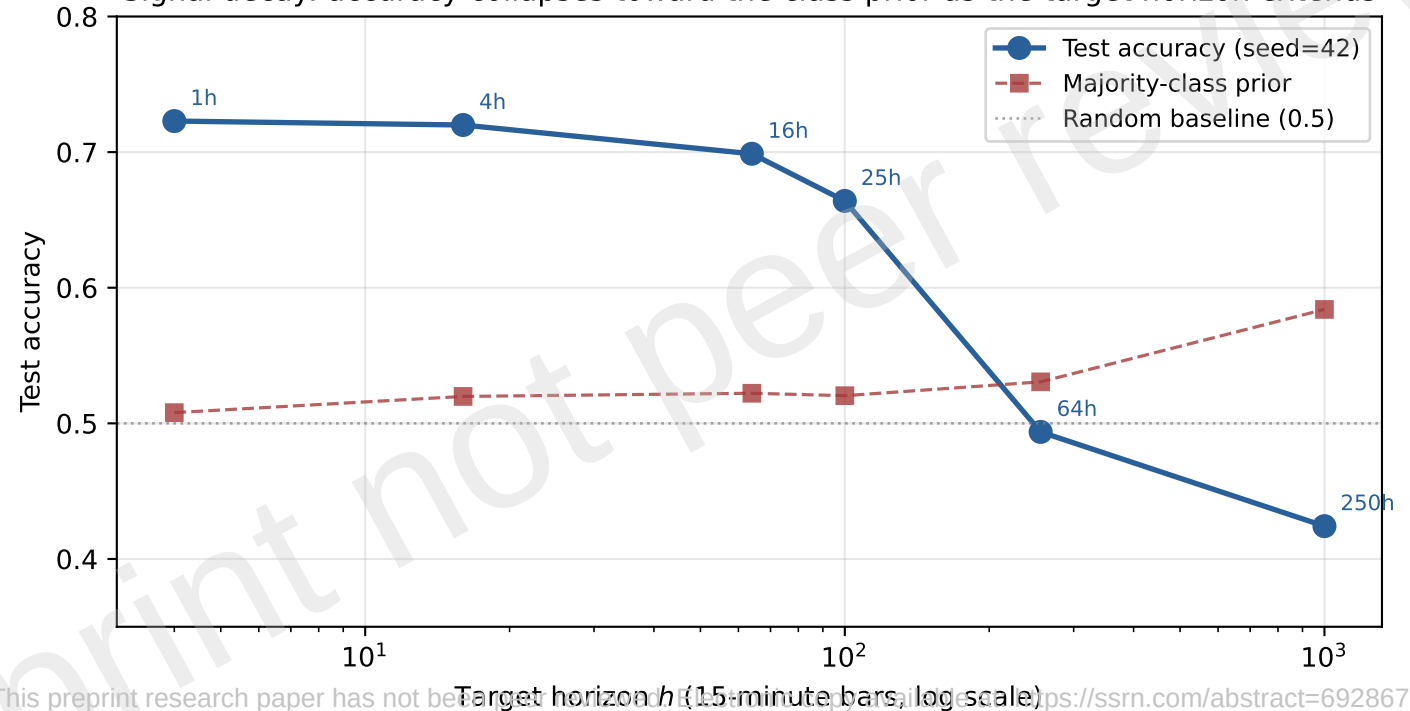
(b) accuracy-based noise vs κ (n=18, Pearson $r=-0.126$, $p=0.617$)



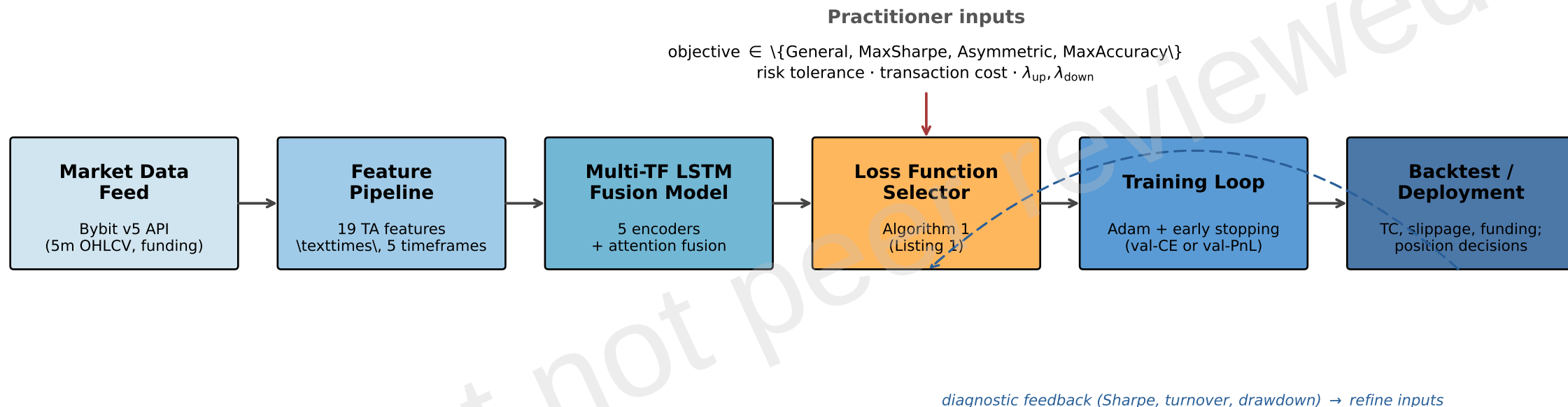
PGL Learning Curves: Early Stopping Interaction



Signal decay: accuracy collapses toward the class prior as the target horizon extends



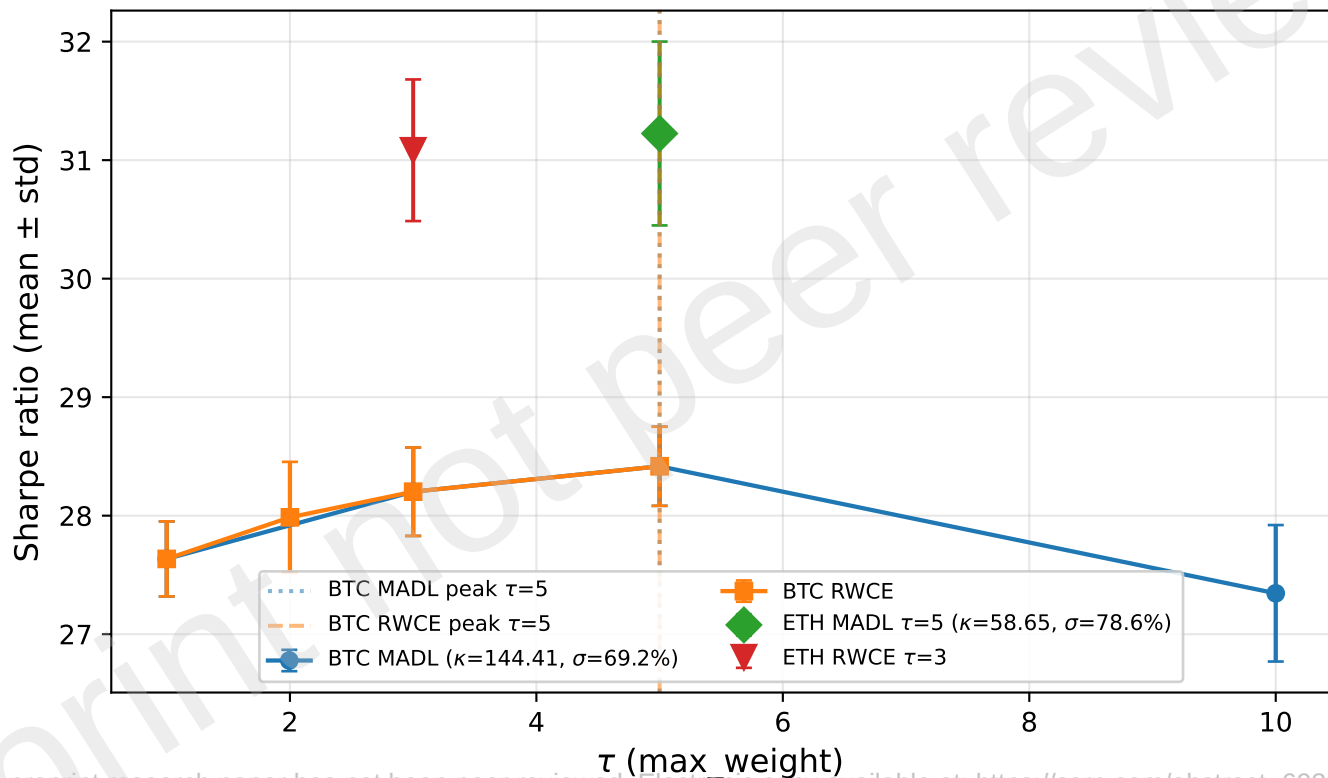
Automated cryptocurrency trading pipeline: the loss-function selector as a pluggable component



Components contributed by this paper:

- Algorithm 1 loss selector (Listing 1)
- Multi-TF fusion model (Section 3)
- Training loop (loss-aware, val-CE / val-PnL options)

Inverted-U of Sharpe vs τ under different market regimes (κ, σ)



Beyond Accuracy: A Systematic Comparison of Profit-Aware Loss Functions for Multi-Timeframe Cryptocurrency Price Prediction

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Abstract

Deep learning for cryptocurrency price prediction is typically trained with cross-entropy, which penalizes all errors equally regardless of financial impact. We compare five profit-aware losses on a multi-timeframe LSTM-fusion backbone (5+ yr BTC, 10 seeds): MADL, Return-Weighted CE, Profit-Guided, Sharpe-ratio, and Asymmetric Directional. MADL and our Return-Weighted CE variant collapse to one family $\mathcal{L}_{RW}(\tau)$ differing only in the clipping bound τ . Return-weighted losses improve per-bar net return from 14.83 to 15.55–15.57 bp (+4.9–5.0%; paired t -test $p < 0.001$ Bonferroni-corrected, $m = 3$) at matched accuracy, with 58% lower cross-seed bp variance. The hourly-annualized Sharpe gain (+5.4–5.5%) is conditional on i.i.d. $\sqrt{8760}$ scaling: daily aggregation gives +0.4–3.1% and a 10-seed Newey–West HAC re-annualization preserves the ranking but is borderline ($p_{\text{Bonf}} = 0.108$ for MADL); Cohen’s $d > 2$ reflects cross-seed variance only. A τ -sweep across three walk-forward regimes shows all $\tau \in [2, 15]$ statistically indistinguishable per fold. A bias–variance argument explains this robustness via an empirical κ – λ coupling: cross-seed Sharpe std (λ proxy) correlates with $\log \kappa$ at

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22 Pearson $r = +0.633$ across 18 cells, and at $r = +0.9995$ (fold-mean check,
 23 $p = 0.020$). We package findings as a decision-support algorithm with a refer-
 24 ence Python implementation mapping trading objective, risk tolerance, and
 25 transaction-cost rate to a concrete loss and hyperparameter choice. Cross-
 26 exchange/cross-asset/walk-forward replications confirm direction robustness;
 27 magnitude is regime-dependent.
 28 *Keywords:* Loss function design, Cryptocurrency prediction,
 29 Multi-timeframe analysis, Deep learning, Trading profitability

30 1. Introduction

31 Deep learning for cryptocurrency price prediction has produced models
 32 with strong classification accuracy via LSTM [1], attention [2], and their
 33 variants [3, 4]. Yet an accuracy–profitability gap persists: 70% directional
 34 accuracy does not guarantee positive returns once transaction costs, slippage,
 35 and heterogeneous return magnitudes are considered [5]. Standard cross-
 36 entropy penalizes a missed 5% move identically to a missed 0.01% move,
 37 motivating profit-aware loss functions [6, 7, 8].

38 Three challenges hinder adoption: **(1) Learning instability**—direct
 39 financial-metric optimization often suffers gradient instability and single-
 40 class collapse [9]. **(2) Lack of systematic comparison.** **(3) Domain**
 41 **specificity**—most losses target equities, yet cryptocurrency returns exhibit
 42 extreme leptokurtosis (BTC 15m simple-return excess kurtosis $\kappa = 162.2$, vs.
 43 $\kappa \approx 5$ –15 for intraday equity returns; Cont 10), positive skew ($\gamma_1 = 1.02$),
 44 and Jarque–Bera $p < 10^{-10}$ —precisely the regime where return-magnitude
 45 weighting should confer the greatest advantage.

We systematically compare five profit-aware losses, and through algebraic analysis identify that MADL and RWCE collapse to a single family $\mathcal{L}_{\text{RW}}(\tau)$ differing only in the clipping bound τ . Our contributions are:

1. **Broad τ -robustness with an empirical mechanism:** a τ -sweep across three walk-forward regimes ($\kappa \in [13, 92]$, $\tau \in \{1, \dots, 15\}$, 63 runs) shows all $\tau \geq 2$ are statistically indistinguishable per fold (only $\tau = 1$ is weaker). Cross-seed Sharpe std (the λ proxy) correlates with $\log \kappa$ at Pearson $r = +0.633$, $p = 0.005$ over 18 cells, while accuracy is regime-invariant—so the tail- and noise-driven shifts of τ^* cancel (Section 5.9, Figure I.11). The primary economic claim is the per-bar net return (+4.9–5.0%, paired- t $p < 0.001$ Bonferroni-corrected, significant under every robustness check); the +5.4–5.5% hourly-annualized Sharpe is corroborative but IID-dependent (Section 4.7, $p_{\text{Bonf}} = 0.108$ for MADL under Newey–West HAC). In our implementation MADL [6]—to which we add clip+renormalize for gradient stability—and RWCE share an identical core differing only in τ .
2. **Dual-metric systematic comparison with statistical rigor:** classification (accuracy, F1, MCC) and trading metrics (Sharpe, max draw-down, per-bar bp) over 10 seeds with paired t -tests, Wilcoxon, Bonferroni and Holm corrections, ablations, transaction-cost sensitivity, Newey–West HAC adjustment, and Probabilistic/Deflated Sharpe Ratios.
3. **Cross-asset, cross-exchange, and temporal validation:** ETH/USDT (Bybit perpetual), BTC/USDT (Binance spot), and 3-fold walk-forward (Page’s trend test $p < 0.001$) all replicate the BTC ranking; failure

71 modes of PGL and Sharpe Loss are diagnosed (Sections 5.7–5.9).

72 **4. Decision support framework** (Algorithm 1): an empirically-grounded
73 selection rule mapping trading objectives, risk tolerance, and cost struc-
74 ture to a recommended loss and hyperparameters, directly integrable
75 into training pipelines.

76 The remainder is organized as follows. Section 2 reviews related work.
77 Section 3 describes methodology. Section 4 presents the experimental setup.
78 Section 5 reports results. Section 6 presents the decision framework. Sec-
79 tion 7 discusses implications. Section 8 concludes.

80 **2. Related Work**

81 **Deep learning for cryptocurrency prediction.** LSTM-based ap-
82 proaches [11, 12], financial-DL surveys [3, 13], and systematic reviews [5]
83 consistently identify a gap between reported accuracy and trading viability
84 under the exclusive use of CE/MSE objectives.

85 **Multi-timeframe analysis** is well-established in technical trading [14]
86 and has been extended to deep learning [15]; we treat the fusion architecture
87 as a fixed backbone and quantify its contribution via a single-TF ablation
88 (Section 5.6.2).

89 **Profit-aware loss functions.** *Return-weighted:* Michańków et al. [6]
90 introduced MADL (CE weighted by price-movement magnitude); we addi-
91 tionally evaluate Return-Weighted CE, shown to be MADL up to clipping
92 (Section 5.6.1). The philosophy connects to meta-labeling [16] but with
93 magnitude integrated into the primary loss. *Profit-guided:* Kar et al. [7]

introduced four such losses (StockLoss + L_2 /Max/Norm) for transformer-based equity portfolios; we evaluate the simpler binary-direction analogue $\mathcal{L}_{CE} - \alpha \cdot \overline{\text{pos}} \cdot \bar{r}$ widely used as a baseline in single-asset settings. *Risk-adjusted*: Moody and Saffell [9] pioneered direct Sharpe optimization [17]; Buehler et al. [18] extended to hedging. *Asymmetric*: Dessain [8] proposed direction-dependent penalties. *RL alternative*: [9, 19, 20] directly optimize returns but face sample-efficiency and stability challenges; modifying only the loss preserves the supervised pipeline’s stability, interpretability, and deployment ease.

No prior study compares these losses within a unified framework on cryptocurrency markets. Table 1 summarizes the taxonomy.

Table 1: Taxonomy of loss functions compared in this study.

Loss Function	Category	Key Mechanism
Cross-Entropy (CE)	Standard	Equal penalty for all errors
MADL [†]	Return-weighted	Weight by $ r_i $, $\tau = 5$
Return-Weighted CE [†]	Return-weighted	Weight by $ r_i $, $\tau = 3$
Profit-Guided	Profit-direct	$CE - \alpha \cdot \text{PnL}$
Sharpe Loss	Risk-adjusted	$-\text{Sharpe}(\text{strategy returns})$
Asymmetric Directional	Asymmetric	Direction-dependent penalty

[†]Same formula in our implementation, differing only in the clipping bound τ (MADL: $\tau = 5$, RWCE: $\tau = 3$); the original MADL of Michańków et al. [6] does not apply weight clipping, so the $\tau = 5$ entry here is the MADL principle as implemented in this paper, not as published.

3. Methodology

3.1. Multi-Timeframe Fusion Framework

The framework processes five timeframes (5m, 15m, 1h, 4h, 1d) in three stages.

Stage 1: Per-timeframe encoding. Each timeframe k has an independent 2-layer LSTM encoder (hidden size $H = 128$, dropout 0.2): $\mathbf{h}^{(k)} = \text{LSTM}_k(\mathbf{X}^{(k)})$, with $\mathbf{X}^{(k)} \in \mathbb{R}^{T \times d_k}$, $T = 48$, $d_k = 19$. Encoders share no parameters.

Stage 2: Scaled dot-product attention fusion. Stack $\{\mathbf{h}^{(k)}\}$ into $\mathbf{H} \in \mathbb{R}^{5 \times H}$ and apply $\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{H})\mathbf{V}$ with learned $W_Q, W_K, W_V \in \mathbb{R}^{H \times H}$. Per-timeframe importance weights $\boldsymbol{\alpha} \in \mathbb{R}^5$ (average over the query dim) yield $\mathbf{h}_{\text{fused}} = \boldsymbol{\alpha}^\top \mathbf{V}$.

Stage 3: Classifier. Linear(128→64)→ReLU→Dropout(0.2)→Linear(64→2) produces binary direction logits. Total parameters: 1.1M.

3.2. Loss Function Formulations

All losses share the interface $\mathcal{L}(\mathbf{z}, y, r)$ with logits $\mathbf{z}$, direction label $y \in \{0, 1\}$, and signed return $r \in \mathbb{R}$. Let $p_i = \text{softmax}(\mathbf{z}_i)_1$, $\text{pos}_i = 2p_i - 1$, and $\overline{|r|}_{\mathcal{B}}$ denote the batch mean absolute return.

Cross-Entropy (baseline): $\mathcal{L}_{\text{CE}} = -N^{-1} \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$.

MADL [6]: $\mathcal{L}_{\text{MADL}} = N^{-1} \sum_i w_i \text{CE}(\mathbf{z}_i, y_i)$ with $w_i = \text{clip}(|r_i| / \overline{|r|}_{\mathcal{B}}, 0.1, 5.0)$, renormalized to unit mean. The original MADL applies neither clipping nor renormalization; we add both for gradient stability, the same recipe canonical for RWCE (Section 5.6.1).

129 **Return-Weighted CE**: same form as MADL with $\tau = 3.0$ (clip to
130 $[0.1, 3.0]$).

131 **Profit-Guided** [7]: $\mathcal{L}_{\text{PG}} = \mathcal{L}_{\text{CE}} - \alpha N^{-1} \sum_i \text{pos}_i r_i$. A warmup schedule
132 linearly ramps α from 0 to target over the first $2w$ epochs ($w = \text{warmup}$,
133 Table 2).

134 **Sharpe Ratio Loss** [9]: $\mathcal{L}_{\text{Sharpe}} = \beta \mathcal{L}_{\text{CE}} + (1 - \beta) \left(-\mathbb{E}[\text{pos} \cdot \tilde{r}] / (\text{std}(\text{pos} \cdot \tilde{r}) + \epsilon) \right)$
135 with $\tilde{r} = r / (|\overline{r}|_{\mathcal{B}} + \epsilon)$, $\epsilon = 10^{-6}$; β ramps from 1.0 to floor 0.5 via
136 warmup.

137 **Asymmetric Directional** [8]: $\mathcal{L}_{\text{Asym}} = N^{-1} \sum_i a_i w_i \text{CE}(\mathbf{z}_i, y_i)$ with $a_i =$
138 $1 + y_i(\lambda_{\text{up}} - 1)(1 - p_i) + (1 - y_i)(\lambda_{\text{down}} - 1)p_i$ and w_i the return-magnitude weight
139 ($\tau = 3$). Symmetric $\lambda_{\text{up}} = \lambda_{\text{down}}$ makes a_i return-independent; genuine non-
140 LRW behavior requires $\lambda_{\text{up}} \neq \lambda_{\text{down}}$.

141 4. Experimental Setup

142 4.1. Dataset

143 BTC/USD [21] Bybit perpetual futures (2019-10 to 2024-12, ~ 5.3 years;
144 552,672 5-minute OHLCV bars, $\sim 99.2\%$ of the calendar maximum 1935 days
145 $\times 288$ bars/day = 557,280; the residual $\sim 0.8\%$ gap is exchange maintenance
146 windows, which we leave as missing rather than forward-filling). Aggregated
147 into five timeframes: 5m, 15m (184,224), 1h (46,056), 4h, 1d. Chronologi-
148 cal split: 70%/15%/15% (train through 2023-06; val through 2024-03; test
149 through 2024-12).

150 4.2. Features

151 19 features per timeframe [22, 14]: *Price* (2: returns, log-returns); *Mov-*
152 *ing averages* (6: SMA-7/25/50/99 ratios, EMA-12/26 ratios); *Oscillators*

153 (4: RSI(14), RSI-norm, MACD-hist, MACD-norm); *Volatility* (5: BB po-
154 sition, ATR(14), ATR-ratio, rolling-vol(10/20)); *Volume* (2: vol-SMA(10),
155 vol-ratio).

156 Target: 1h-horizon direction, $y_i = \mathbb{I}[p_{t+4} > p_t]$ on the 15m timeframe.
157 Shorter horizons are noise-dominated; longer horizons reduce tradeable sig-
158 nals. Label distribution: 50.7% up / 49.3% down (balanced).

159 4.3. Data Integrity and Look-Ahead Bias Prevention

160 We take explicit precautions against data leakage. **Feature normaliza-**
161 **tion:** technical indicators use only past data; StandardScaler is fit on train-
162 ing only. **Indicator right-edge convention:** all rolling indicators (SMA,
163 EMA, RSI, MACD, ATR, BB, rolling vol, volume SMA) are computed via
164 **pandas** causal rolling/EWM operators; for a prediction at index t we use the
165 feature row at $t - 1$. **Timeframe alignment:** higher-TF bars are indexed
166 via right-bounded binary search, using only the most recent closed bar be-
167 fore t (worst case: 23h59m stale daily bar). The target $y_i = \mathbb{I}[p_{t+4} > p_t]$
168 references a strictly future close; splits are chronological.

169 **Empirical leakage control.** *Target shuffling:* training with permuted
170 labels yields test accuracy $50.79 \pm 0.00\%$ (the up-class proportion) vs. $72.20 \pm$
171 0.39% under normal labels; the shuffled model collapses to a single-class
172 predictor (Table D.20), ruling out above-chance leakage. *Target-horizon shift*
173 (single seed): retraining with $\mathbb{I}[p_{t+h} > p_t]$ for $h \in \{4, 16, 64, 100, 256, 1000\}$
174 (Figure 1) yields monotonic decay from 72.3% to 42.4%, within the $\approx 50\% \pm$
175 8% prior range; horizon-invariant leakage would hold accuracy near 72%.
176 This rules out horizon-invariant but not horizon-specific paths; a non-fusion-
177 backbone shift test is left for future work.

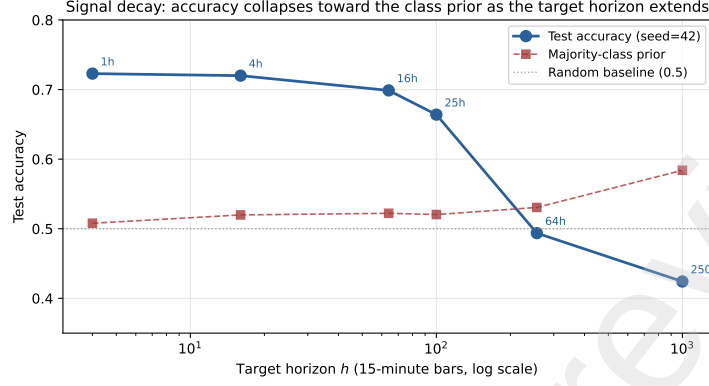


Figure 1: Signal decay across target horizons. The lower x -axis is the horizon in 15-minute bars $h \in \{4, 16, 64, 100, 256, 1000\}$; the upper x -axis is the corresponding horizon in hours $\{1, 4, 16, 25, 64, 250\}$. Single seed (seed 42); the smooth monotonic decay from 72% to the majority-class prior rules out horizon-invariant leakage.

178 4.4. Model Configuration

179 2-layer LSTM encoders ($H = 128$, dropout 0.2), attention fusion, Adam
 180 (lr 0.001, batch 64), early stopping (patience 10, max 100 epochs).

181 4.5. Loss Function Hyperparameters

182 Table 2. Defaults follow original publications. MADL $\tau = 5$ vs. RWCE
 183 $\tau = 3$ because the original MADL [6] does not clip; we use τ in mathematical
 184 expressions and the code-level `max_weight` in Listing 1.

185 4.6. Evaluation Protocol

186 Each loss is trained with 10 seeds (42–51). Cohen’s d for Sharpe vs.
 187 CE: MADL 2.63, RWCE 2.51, Asymmetric 2.09 (95% CIs exclude zero, Ta-
 188 ble E.21). Tests: paired t -test (primary, $n = 10$ seed-level means) with

Table 2: Loss function hyperparameters.

Loss Function	Hyperparameters
Cross-Entropy	(none)
MADL	$\tau = 5.0$
Profit-Guided	$\alpha = 0.5$, warmup = 5 epochs
Sharpe Loss	$\epsilon = 10^{-6}$, warmup = 5, CE floor = 0.5
Return-Weighted CE	$\tau = 3.0$
Asymmetric Directional	$\lambda_{\text{up}} = 1.5$, $\lambda_{\text{down}} = 1.5$, $\tau_{\text{mag}} = 3.0$

189 Bonferroni/Holm corrections, Wilcoxon signed-rank [23], and Friedman (Ap-
 190 pendix Appendix F only; Demšar 24’s protocol is methodologically inappro-
 191 priate for single-dataset use).

192 *Caveat on Cohen’s d .* The across-seed std captures training stochasticity
 193 only, not market-level variance. Large $d > 2$ reflects a Sharpe gap much larger
 194 than seed-level noise, not robustness across markets; the latter is addressed
 195 via walk-forward (Section 5.9), ETH (Section 5.8), and Binance (Section 5.7).

196 4.7. Trading Simulation

197 Backtest: \$10,000 notional, transaction cost 0.1% per trade (round-trip
 198 0.2%), slippage 0.05%. Simple (non-compound) returns with full-size posi-
 199 tions on every signal. Soft position $\text{pos}_i = 2p_i - 1$ is used only inside loss
 200 computation (PGL, Sharpe Loss); trading uses hard decisions ($p > 0.5 \Rightarrow +1$,
 201 else -1). Section 5.6.3 varies fees across five levels.

202 Sharpe is annualized as $\text{Sharpe}_{\text{annual}} = \text{Sharpe}_{\text{hourly}} \times \sqrt{8,760}$ (24/7 crypto).
 203 Hourly 0.27–0.31 $\Rightarrow$ annualized 25–29. The $\sqrt{T}$ scaling assumes i.i.d. returns;

we provide three robustness checks. (i) Daily aggregation via $\sqrt{365}$ (Table 5): 7–16% lower than the hourly counterpart. (ii) Newey–West HAC [25] (Bartlett, $L = \lfloor 4(T/100)^{2/9} \rfloor = 13$): for CE the $1.65\times$ inflator reduces 27.07 to 16.37 (Table 3); MADL > RWCE > CE ranking preserved, paired- t borderline ($p_{\text{Bonf}} = 0.108$ for MADL). (iii) PSR/DSR [26] with SR^* for $N = 6$ (Table 4): both = 1.000 at all 30 cells. Per-bar bp and daily-aggregated Sharpe are the primary evidence and do not depend on $\sqrt{8760}$.

Table 3: Newey–West HAC Sharpe (10 seeds, Bartlett kernel $L = 13$, seeds 42–51). “SE ratio”=NW-to-IID SE inflator. p -value: paired t -test vs. CE; Bonferroni $m = 2$ gives 0.108 (MADL), 0.444 (RWCE). IID Sharpe here is computed from pooled hourly mean/std and differs by ≤ 0.5 from Table 7 (simulator’s portfolio routine).

Loss	Sharpe IID	Sharpe NW	$\overline{\text{SE}_{\text{NW}}/\text{SE}_{\text{IID}}}$	p vs CE (NW)
CE	27.07 ± 0.72	16.37 ± 0.38	1.65	—
MADL	28.39 ± 0.43	16.73 ± 0.28	1.70	0.054
RWCE	28.02 ± 0.54	16.64 ± 0.35	1.68	0.222

Cumulative ROI is omitted to avoid misinterpretation; only per-bar average return (~ 15 bp net of costs, $n_{\text{bar}} = 23,040$) and annualized Sharpe are retained.

4.8. Reproducibility

All experiments ran on a single NVIDIA RTX 5070 Ti (16 GB VRAM) with CUDA 12.8 and PyTorch 2.7. Each run takes 2–3 min/seed; total budget for the main experiment ($6 \text{ losses} \times 10 \text{ seeds} = 60 \text{ runs}$) is ~ 2.5 h. Random seeds are fixed at four levels (`random`, `numpy`, `torch`, `cuda`) with

Table 4: PSR(0) and DSR [26] on hourly per-bar net returns ($T = 23,040$). PSR uses empirical skew/raw-kurt; DSR uses $SR^* = \sigma_{SR} \cdot ((1 - \gamma_E)\Phi^{-1}(1 - 1/N) + \gamma_E\Phi^{-1}(1 - 1/(Ne)))$ with $N = 6$, $\sigma_{SR} = 0.0085 \Rightarrow SR^* = 0.0110$. All 30 (loss, seed) cells reach 1.000, ruling out null Sharpe and selection bias.

Loss	SR_h (mean $\pm$ std)	PSR(0)	DSR
CE	0.2893 ± 0.0077	1.000	1.000
MADL	0.3034 ± 0.0046	1.000	1.000
RWCE	0.2994 ± 0.0058	1.000	1.000

Table 5: Daily-aggregated Sharpe (mean $\pm$ std, 3 seeds, $\sqrt{365}$ annualization). 3-seed count due to per-bar-return artefact availability. Daily values are 7–16% lower than hourly over the same seeds. Sharpe Loss flips from -5.0% hourly to $+1.8\%$ daily (autocorrelation under β -warmup); Asymmetric’s -1.7% is the only sign reversal in the return-weighted trio (see Section 6).

Loss Function	Daily Sharpe (annual)	Δ vs CE
CE	23.89 ± 1.10	—
MADL	23.99 ± 0.14	+0.4%
Profit-Guided	24.15 ± 1.04	+1.1%
Sharpe	24.31 ± 0.87	+1.8%
RWCE	24.63 ± 0.30	+3.1%
Asymmetric	23.49 ± 1.29	-1.7%

219 `cuda.deterministic = True`; seeds 42–51 are used. A parallel widely-
 220 spaced seed set (`results/nonconsecutive_seeds.json`) is used for Monte-
 221 Carlo-independent verification. BTC/USD data is from the Bybit v5 API
 222 (USDT perpetual, 2019-10-01 to 2024-12-31, downloaded 2026-01-15). Source
 223 code, trained configurations, and data download scripts are available from
 224 the author on reasonable request.

225 5. Results

226 5.1. Overall Comparison

227 Table 6 reports main results across six losses (10 seeds), organized as
 228 classification (accuracy, F1, MCC) and trading (Sharpe, per-bar bp, max
 229 drawdown).

Table 6: Classification performance (mean $\pm$ std over 10 seeds). Bold indicates best, underline indicates second best. */**/**: $p < 0.05/0.01/0.001$ vs. CE (paired t -test); absence of marker indicates $p \geq 0.05$.

Loss Function	Accuracy	F1	MCC
CE	<u>0.7228 $\pm$ 0.0032</u>	<u>0.7226 $\pm$ 0.0032</u>	0.4471 $\pm$ 0.0064
MADL	0.7218 $\pm$ 0.0033	0.7215 $\pm$ 0.0036	0.4451 $\pm$ 0.0061
Profit-Guided	0.7198 $\pm$ 0.0033*	0.7193 $\pm$ 0.0037*	0.4425 $\pm$ 0.0048
Sharpe	0.7084 $\pm$ 0.0036***	0.7078 $\pm$ 0.0040***	0.4197 $\pm$ 0.0066***
RWCE	0.7231 $\pm$ 0.0027	0.7226 $\pm$ 0.0031	0.4489 $\pm$ 0.0044
Asymmetric	0.7224 $\pm$ 0.0024	0.7220 $\pm$ 0.0025	<u>0.4472 $\pm$ 0.0054</u>

Table 7: Trading performance (mean $\pm$ std over 10 seeds). The primary economic metric is **Avg/bar (bp)**, the per-bar mean strategy return net of transaction costs (Avg/bar = $\text{ROI}\% \times 10000 / (100 \times n_{\text{bar}})$ with $n_{\text{bar}} = 23,040$ evaluated 15-minute bars). Annualized Sharpe uses $\sqrt{8760}$ scaling and assumes i.i.d. returns; daily-aggregated and Newey–West HAC adjusted versions appear in Tables 5 and 3. Cumulative non-compound ROI is omitted to avoid misinterpretation.

Loss Function	Sharpe (annual)	Avg/bar (bp)	Max DD (%)	Δ Sharpe vs CE
CE	27.07 ± 0.72	14.83	-2.61 ± 0.47	—
MADL	<u>$28.53 \pm 0.30^{***}$</u>	<u>15.55</u>	-2.34 ± 0.22	+5.4%
Profit-Guided	26.73 ± 0.63	14.66	-2.65 ± 0.44	−1.3%
Sharpe (defaults)	$25.73 \pm 1.17^{**}$	14.14	-2.65 ± 0.58	−5.0%
<i>Sharpe (best-tuned)[‡]</i>	<i>28.00 (n = 3)</i>	<i>15.36</i>	—	+3.4%
RWCE	$28.52 \pm 0.38^{***}$	15.55	-2.51 ± 0.26	+5.4%
Asymmetric	$28.56 \pm 0.69^{***}$	15.57	<u>-2.51 ± 0.35</u>	+5.5%

[‡]Best Sharpe Loss configuration ($\beta = 0.3$, warmup = 0; Appendix Appendix C).

Reported on 3 seeds; the Δ Sharpe shown (+3.4%) is referenced to this table’s 10-seed CE baseline of 27.07. Appendix Appendix C reports the same configuration as +6.0% versus its same-3-seed CE baseline of 26.42; the two figures describe the same underlying run with different baselines.

230 5.1.1. Classification Performance

231 All losses achieve directional accuracy 70.8–72.3%. CE (72.28%), MADL
232 (72.18%), RWCE (72.31%, highest), and Asymmetric (72.24%) are statisti-
233 cally indistinguishable ($p > 0.05$); Sharpe Loss is significantly lower (70.84%,
234 $p < 0.001$); PGL shows a modest drop (71.98%, $p < 0.05$). Direct financial
235 optimization can degrade accuracy when its loss landscape conflicts with the

236 CE landscape.

237 5.1.2. Trading Performance

238 Trading performance reveals a different pattern: MADL, RWCE, and
239 Asymmetric significantly outperform CE on Sharpe ratio ($p < 0.001$ paired
240 t -test; Table 8 for corrections):

- 241 • **Sharpe ratio:** Asymmetric (28.56, +5.5%), MADL (28.53, +5.4%),
242 and RWCE (28.52, +5.4%) are statistically indistinguishable (pairwise
243 $p > 0.5$; cross-seed std 0.30–0.69) and collectively ~ 5.4 – 5.5% above CE
244 (27.07). Absolute annualized Sharpe values (25–29) are inflated by $\sqrt{T}$
245 scaling (see Section 4.7).
- 246 • **Per-bar net return:** 15.55–15.57 bp for the return-weighted trio vs.
247 14.83 bp for CE (+0.72–0.74 bp/bar, a ~ 4.9 – 5.0% relative gain).
- 248 • **Maximum drawdown:** MADL achieves the shallowest drawdown
249 (-2.34% vs. CE -2.61%); this tiebreaker underlies the MAXSHARPE/AGGRESSIVE
250 recommendation in Algorithm 1.
- 251 • **Variance:** the cross-seed per-bar-bp std for MADL is 58% lower than
252 for CE (Avg/bar std scales as ROI std/ n_{bar}). MADL has the lowest
253 cross-seed Sharpe std in the top tier (0.30 vs. RWCE 0.38, Asymmetric
254 0.69); RWCE is the second-lowest and is paired with the highest ac-
255 curacy (72.31%), underlying the CONSERVATIVE/GENERAL branches
256 that prioritise accuracy retention together with low cross-seed variance.

257 PGL and Sharpe Loss underperform CE at literature defaults; Sharpe
258 Loss recovers under grid search (Appendix Appendix C), so its underperfor-

259 mance reflects tuning cost not fundamental incompatibility. The +4.9–5.0%
 260 per-bar net-return improvement for the return-weighted trio is significant
 261 at $p < 0.001$ Bonferroni-corrected under *every* robustness check (paired t ,
 262 Wilcoxon, ANCOVA, PSR/DSR), and is therefore the primary economic
 263 claim; the +5.4–5.5% hourly-annualized Sharpe is corroborative but condi-
 264 tional on the $\sqrt{8760}$ i.i.d. assumption (Section 4.7 for alternatives).

265 5.2. Statistical Significance

266 Figure 2 shows pairwise paired- t p -values; Table 8 reports the exact values
 267 with multiple-comparison corrections for Sharpe vs. CE.

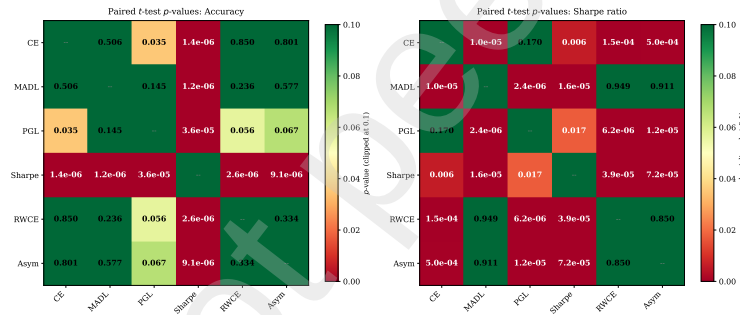


Figure 2: Pairwise paired t -test p -values for accuracy (left) and Sharpe ratio (right).

Table 8: Pairwise comparison vs. CE baseline with multiple comparison corrections (Sharpe ratio, 10 seeds).

Loss	ΔSharpe	p_{raw}	p_{Bonf}	p_{Holm}	$p_{\text{Wil,Bonf}}$
MADL	+1.46	<0.001	<0.001	<0.001	0.0098
ProfitGuided	-0.34	0.1699	0.8495	0.1699	1.0000
Sharpe	-1.34	0.0061	0.0306*	0.0123*	0.0488*
ReturnWeightedCE	+1.45	<0.001	<0.001	<0.001	0.0098
AsymmetricDirectional	+1.48	<0.001	0.0025	0.0015	0.0098

Three tiers emerge: on *accuracy*, Tier 1 = {CE, MADL, RWCE, Asym} (no pairwise differences), Tier 2 = {PGL}, Tier 3 = {Sharpe Loss}. On *Sharpe*, Tier 1 = {MADL, RWCE, Asym} (all above CE, $p < 0.001$), Tier 2 = {CE, PGL}, Tier 3 = {Sharpe Loss} (borderline $p_{\text{Bonf}} = 0.031$).

Multiple comparison correction. Following Harvey et al. [27], three primary comparisons (MADL/RWCE/Asymmetric vs. CE on Sharpe) are pre-specified; Bonferroni at $m = 3$ yields $p_{\text{Bonf}} < 0.001$ for MADL and RWCE and $p_{\text{Bonf}} \leq 0.002$ for Asymmetric. For full transparency, Table 8 reports the more conservative all-pairs Bonferroni correction at $m = 5$ (all five non-CE losses), under which Asymmetric remains significant at $p_{\text{Bonf}} = 0.003$ and the qualitative conclusions are unchanged. Holm–Bonferroni and Wilcoxon signed-rank (Bonferroni) corroborate. The Friedman–Nemenyi analysis (Table F.22) is reported in Appendix Appendix F only; see Section 4.6 for the methodological caveat.

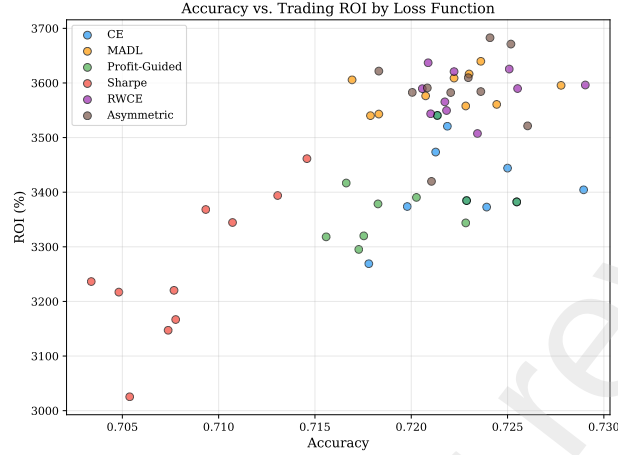


Figure 3: Accuracy vs. cumulative non-compound ROI per seed and loss. Return-weighted losses shift the cloud upward at matched accuracy. The y -axis uses ROI for visual scale only; for the primary per-bar bp metric divide ROI by $n_{\text{bar}} = 23,040$.

5.3. Accuracy-Profitability Trade-off

An ANCOVA model $\text{ROI} \sim \text{accuracy} + C(\text{loss})$ on the 60 observations (Figure 3) confirms this statistically: loss factor $F(5, 53) = 13.99$, $p = 1.0 \times 10^{-8}$, $R^2 = 0.78$. Accuracy-adjusted shifts vs. CE: MADL +1.78 ($p = 1.5 \times 10^{-6}$), RWCE +1.63 ($p = 7.4 \times 10^{-6}$), Asymmetric +1.74 ($p = 2.2 \times 10^{-6}$); PGL/Sharpe Loss not significant. Companion ANCOVA on Sharpe: $F = 12.96$, $p = 3.0 \times 10^{-8}$; interaction term borderline ($p = 0.055$ ROI, 0.064 Sharpe). The upward shift is driven by better accuracy-to-ROI conversion, not accuracy differences. Cross-seed boxplots (Figure H.10, Appendix Appendix H) show tighter distributions for return-weighted variants.

5.4. Market Condition Analysis

We classify each test bar by trailing 24h return (ex-ante): *up* ($r > 2\%$, 4,526 bars), *down* ($r < -2\%$, 3,681), *flat* ($|r| \leq 2\%$, 14,833). The total

295 $n_{\text{bar}} = 23,040$ equals the test set size (15% of $184,224 \approx 27,634$) minus the 1d-
 296 timeframe warmup of $48 \times 96 = 4,608$ bars plus a ~ 14 -bar residual absorbed
 297 by the prediction horizon and chronological-split alignment. Table 9 reports
 298 per-regime metrics (10 seeds).

Table 9: Performance by market condition (mean over 10 seeds). Bold indicates best Sharpe per condition.

	Up		Down		Flat	
Loss	Acc	Sharpe	Acc	Sharpe	Acc	Sharpe
CE	71.8%	29.6	72.0%	29.6	72.5%	25.4
MADL	71.8%	30.8	72.1%	31.3	72.3%	26.9
RWCE	71.9%	31.0	72.3%	31.3	72.4%	26.8
Asymmetric	71.7%	30.7	72.3%	31.2	72.4%	27.0
ProfitGuided	71.7%	29.0	71.5%	29.1	72.2%	25.1
SharpeLoss	70.9%	27.9	69.9%	26.8	71.0%	24.6

299 Return-weighted losses outperform CE in all three regimes, with the
 300 largest gap in **down periods (MADL 31.3 vs. CE 29.6, +5.7%)**—the
 301 regime where correct directional prediction is most financially consequen-
 302 tial. Applying volatility-proportional slippage inflators ($\alpha_{\text{down}} = 1.411$) to
 303 the loss-by-cost slope from Table 12 widens the down-period MADL–CE gap
 304 to +6.12% (Sharpe 29.69 vs. 27.98), as MADL trades $\sim 8.4\%$ less frequently.

305 5.5. Equity Curve

306 Figure 4 shows non-compound equity curves and drawdowns for CE,
 307 MADL, RWCE (seed 42, \$10,000 initial, 0.1% fees, 0.05% slippage). MADL/RWCE

308 achieve higher terminal equity and shallower drawdowns. The y -axis is the
 309 arithmetic sum of per-bar strategy returns over the 23,040 test bars and
 310 is a model-vs.-model diagnostic, not a realizable portfolio; the full-notional
 311 compound version exhibits non-physical magnitudes reflecting the degenerate
 312 sizing rule (see Section 9 for realistic deployment).

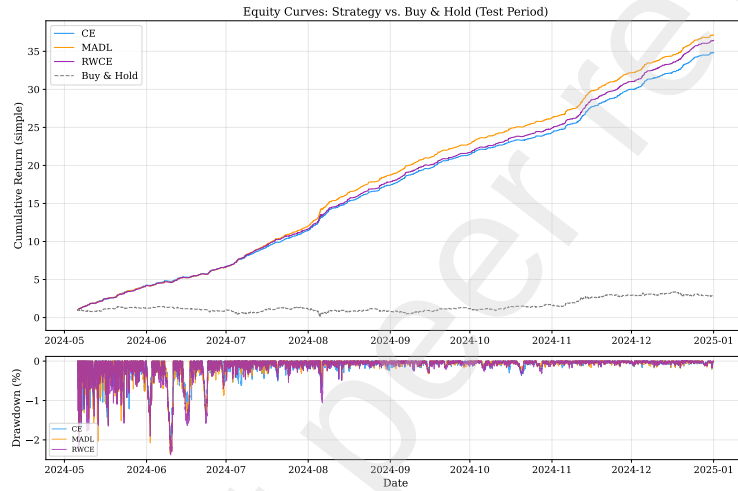


Figure 4: **Methodological diagnostic, not a portfolio simulation.** Non-compound equity (top) and drawdown (bottom). The y -axis is the arithmetic sum of per-bar strategy returns; the apparent magnitude is an artifact of full-size summation across $n_{\text{bar}} = 23,040$ test bars.

313 5.6. Ablation Studies

314 5.6.1. Unified Return-Weighting Family and Clipping Sensitivity

315 The `max_weight` hyperparameter (hereafter τ) controls the upper bound
316 on return-magnitude weights, preventing outlier returns from dominating the
317 loss.

318 **Unified formulation (a structural integration, not a finding).**
319 MADL and RWCE share an identical core formula and differ only in τ
320 (MADL 5, RWCE 3); treating them as a single family $\mathcal{L}_{\text{RW}}(\tau)$ lets us state
321 the τ -sensitivity result on a single axis:

$$\mathcal{L}_{\text{RW}}(\tau) = \frac{1}{N} \sum_{i=1}^N \tilde{w}_i(\tau) \cdot \text{CE}(\mathbf{z}_i, y_i), \quad w_i(\tau) = \text{clip}\left(\frac{|r_i|}{|r|_{\mathcal{B}}}, 0.1, \tau\right), \quad \tilde{w}_i = \frac{w_i}{\bar{w}}. \quad (1)$$

322 Performance differences in Tables 6–7 are entirely attributable to τ , not to
323 distinct loss formulations: practitioners need only tune a single continuous
324 parameter. Under symmetric $\lambda_{\text{up}} = \lambda_{\text{down}}$, $\mathcal{L}_{\text{Asym}} = a_i \cdot \mathcal{L}_{\text{RW}}(\tau = 3)$ with a_i
325 return-independent; the genuinely non-LRW design space is {PGL, Sharpe
326 Loss, Asymmetric with $\lambda_{\text{up}} \neq \lambda_{\text{down}}$ }.

327 We evaluate $\tau \in \{1, 2, 3, 5, 10\}$ over 3 seeds each.

Table 10: Ablation study: effect of the clipping bound τ (code identifier: `max_weight`) on Sharpe ratio and accuracy for the return-weighted CE family (mean $\pm$ std over 3 seeds). MADL and RWCE share the same core formula (see text); rows with matching τ are therefore identical, confirming that the only operational difference between MADL and RWCE in our implementation is the default τ .

Label	τ	Accuracy (%)	Sharpe	ROI (%)	MaxDD (%)
—	1	72.05 $\pm$ 0.36	27.6 $\pm$ 0.3	3480 $\pm$ 41	−2.68
RWCE	2	72.51 $\pm$ 0.11	28.0 $\pm$ 0.5	3526 $\pm$ 48	−2.54
RWCE†	3	72.33 $\pm$ 0.23	28.2 $\pm$ 0.4	3547 $\pm$ 41	−2.43
MADL†	5	72.12 $\pm$ 0.29	28.4 $\pm$ 0.3	3572 $\pm$ 40	−2.38
MADL	10	71.65 $\pm$ 0.12	27.3 $\pm$ 0.6	3452 $\pm$ 65	−2.57

† denotes the default value used in the main experiments.

Figure 5 visualizes the Sharpe ratio sensitivity. Performance increases monotonically from $\tau = 1$ (Sharpe 27.6, comparable to CE) to $\tau = 5$ (Sharpe 28.4), then declines sharply at $\tau = 10$ (Sharpe 27.3, accuracy drops to 71.7%). This inverted-U pattern suggests that moderate return weighting ($\tau \in [3, 5]$) strikes the optimal balance between emphasizing large moves and avoiding outlier-driven gradient noise. The RWCE default ($\tau = 3$, Sharpe 28.2) provides a conservative choice, while the MADL default ($\tau = 5$, Sharpe 28.4) maximizes risk-adjusted returns at a slight accuracy cost.

τ^* and market tail properties. BTC ($\kappa_{\log} = 144.4$, $\sigma = 69.2\%$) and ETH ($\kappa_{\log} = 58.6$, $\sigma = 78.6\%$) are both strongly leptokurtic. The weight saturates above $\tau \cdot \overline{|r|}_B$, so moderate τ up-weights informative mid-magnitude returns while attenuating the tail. As shown in Figure 6, both

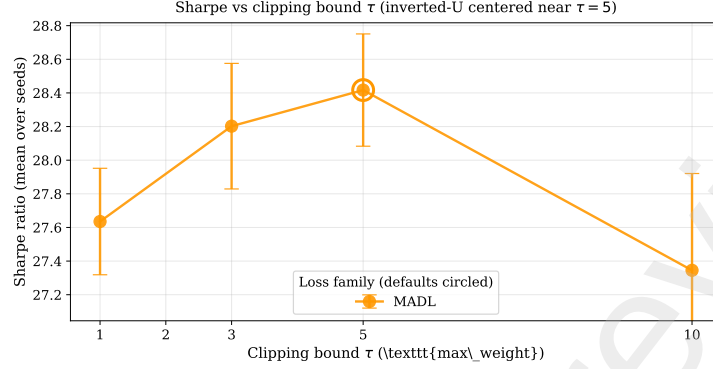


Figure 5: Effect of τ (the weight-clipping bound; coded as `max_weight` in `custom_losses.py`) on Sharpe ratio for MADL and RWCE. Orange bars indicate the default values used in the main experiments.

340 peak in $\tau \in [3, 5]$ at the main-experiment grid; BTC MADL ($\tau = 5$) and
 341 RWCE ($\tau = 3$) match within cross-seed noise, and ETH MADL (31.22) and
 342 RWCE (31.08) differ by 0.14 ($< \text{std}$). The $[3, 5]$ range is a plateau, widened to
 343 $[2, 15]$ within each walk-forward regime (Section 5.9). Extension to additional
 344 assets is future work.

345 5.6.2. Architecture Ablation: Single-Timeframe Baseline

346 To isolate the contribution of multi-timeframe fusion, we retrain the same
 347 losses on the 15-minute timeframe only (the attention block receives a single
 348 token); all other elements are held fixed. Table 11 reports 3-seed means.

349 Accuracy collapses from 71–72% to 50–53%, near the class prior; Sharpe
 350 ratios become uniformly negative (-5.8 to -3.1). The loss-function compari-
 351 son is therefore conditional on multi-timeframe fusion supplying a non-trivial
 352 predictor. We do not re-tune the single-timeframe baseline; the ablation
 353 isolates the fusion contribution under a fixed architecture (prior single-TF

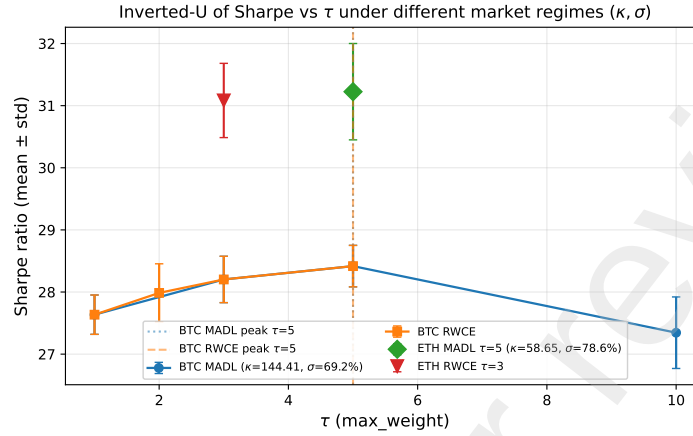


Figure 6: Sharpe vs. τ for BTC and ETH. Both show an inverted-U with a plateau in $\tau \in [3, 5]$ despite $\kappa_{\text{BTC}}/\kappa_{\text{ETH}} > 2$.

Table 11: Single-timeframe (15m only) ablation: 3 seeds per loss function. Accuracy collapses to the class-prior baseline (~ 50 – 53%) compared to the $\sim 72\%$ of the multi-timeframe setup (Table 6); all backtests lose money. The +values in the Δ column do not indicate profits, only smaller absolute losses.

Loss Function	Accuracy (%)	Sharpe	ROI (%)	Δ vs CE Sharpe
CrossEntropyLoss	52.33 ± 0.10	-5.77 ± 0.79	-767.1 ± 105.4	—
MADL	50.26 ± 0.57	-3.35 ± 4.08	-445.4 ± 540.6	+2.42
RWCE	50.64 ± 0.15	-3.13 ± 4.04	-416.9 ± 535.3	+2.63
AsymmetricDirectionalLoss	50.83 ± 0.26	-5.69 ± 0.96	-755.8 ± 128.0	+0.08
ProfitGuidedLoss	52.75 ± 0.39	-5.68 ± 1.16	-756.4 ± 156.4	+0.09
SharpeLoss	51.53 ± 0.61	-3.83 ± 0.36	-506.3 ± 48.3	+1.94

354 baselines [11, 12] reach 60–65% under separately-optimized configurations).

355 5.6.3. Transaction Cost Sensitivity

356 We evaluate CE, MADL, RWCE at five cost levels (0.05%, Bybit taker
357 0.055%, 0.1%, 0.2%, 0.5%).

Table 12: Transaction cost sensitivity analysis (mean over 3 seeds). Bybit taker fee is 0.055%.

Cost Level	Loss	ROI (%)	Sharpe	Total Cost (%)
0.05%	CE	4269.5	35.1	924.2
0.05%	MADL	4417.7	36.5	846.2
0.05%	RWCE	4407.8	36.4	860.8
Bybit taker	CE	4146.3	33.9	1047.5
Bybit taker	MADL	4304.9	35.4	959.0
Bybit taker	RWCE	4293.0	35.3	975.6
0.1% (baseline)	CE	3345.3	26.4	1848.5
0.1% (baseline)	MADL	3571.5	28.4	1692.4
0.1% (baseline)	RWCE	3547.0	28.2	1721.7
0.2%	CE	1496.9	10.6	3696.9
0.2%	MADL	1879.2	13.4	3384.7
0.2%	RWCE	1825.3	13.0	3443.3
0.5%	CE	-4048.5	-19.0	9242.2
0.5%	MADL	-3197.9	-15.4	8461.8
0.5%	RWCE	-3339.6	-16.0	8608.2

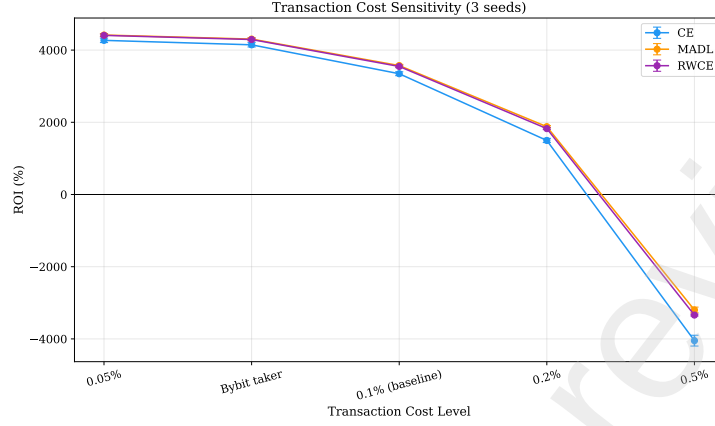


Figure 7: Transaction cost sensitivity. The ranking $\text{MADL} \approx \text{RWCE} > \text{CE}$ is preserved across all cost levels.

Figure 7 visualizes the cost sensitivity: the ranking $\text{MADL} \approx \text{RWCE} > \text{CE}$ holds across all costs. At Bybit taker fees (0.055%) all three are highly profitable (MADL 35.4, CE 33.9); at 0.2% ordering is preserved (MADL 13.4, CE 10.6); at 0.5% all strategies are unprofitable but MADL still beats CE by 3.6 Sharpe points. Return-weighted losses incur $\sim 8\%$ lower total costs than CE (MADL -8.4% , RWCE -6.9% at the 0.1% baseline; computed from Table 12), indicating fewer unnecessary position changes on low-signal bars.

5.7. Cross-Exchange Validation: BTC/USDT Spot on Binance

We replicate on Binance BTC/USDT *spot* (2023-07 to 2024-12, 158k 5-minute bars) with otherwise identical pipeline. Six losses $\times$ 3 seeds (Table 13).

Sharpe ranking: RWCE (29.56) $>$ Asymmetric (28.57) $\approx$ MADL (28.47) $>$ CE (27.86) $\approx$ Sharpe (27.82) $\approx$ PGL (27.77). The family-vs.-non-family distinction reproduces under a different exchange, instrument class, and 18-

Table 13: Binance BTC/USDT spot replication (3 seeds; 2023-07 to 2024-12). Return-weighted losses again dominate CE; PGL/Sharpe Loss remain within noise.

Loss Function	Accuracy (%)	Sharpe	ROI (%)
CrossEntropyLoss	70.56 ± 0.53	27.86 ± 0.38	539.5 ± 5.0
MADL	69.13 ± 0.80	28.47 ± 1.16	549.5 ± 19.6
RWCE	70.18 ± 0.15	29.56 ± 0.91	568.4 ± 16.2
AsymmetricDirectionalLoss	69.74 ± 0.43	28.57 ± 1.10	550.4 ± 19.2
ProfitGuidedLoss	70.43 ± 0.60	27.77 ± 0.58	538.3 ± 8.6
SharpeLoss	68.96 ± 1.13	27.82 ± 0.53	539.3 ± 8.1

month window. The MADL-vs.-CE Sharpe gap shrinks from +5.4% on Bybit perpetual to +2.2% on Binance spot (bull-dominated): the Bull walk-forward fold (Section 5.9) likewise shows the smallest within-fold spread, as trending bull regimes leave less variance for return-weighting to exploit. The direction of all comparisons reproduces; magnitude is regime-conditional.

5.8. Cross-Asset Validation: ETH/USDT

We replicate on ETH/USDT (Bybit perpetual, 2021-03-15 to 2024-12-31, ~400k bars) through the identical pipeline.

Table 14: ETH/USDT validation: loss function comparison (mean $\pm$ std over 10 seeds).
Format matches Table 7 for direct comparison with BTC results.

Loss Function	Accuracy	Sharpe (annual)	ROI (%)	Max DD (%)
Asymmetric	0.7219 ± 0.0025	31.13 ± 0.70	3378 ± 66	-4.86 ± 0.58
CE	0.7232 ± 0.0013	30.45 ± 0.63	3310 ± 61	-5.05 ± 0.46
MADL	0.7204 ± 0.0019	31.22 ± 0.78	3386 ± 72	-5.05 ± 0.79
Profit-Guided	0.7222 ± 0.0021	30.36 ± 0.71	3300 ± 68	-5.12 ± 0.46
RWCE	0.7223 ± 0.0017	31.08 ± 0.60	3372 ± 56	-5.05 ± 0.80
Sharpe	0.7152 ± 0.0018	29.27 ± 0.57	3193 ± 55	-6.30 ± 1.71

Sharpe ranking matches BTC: MADL (31.22), Asymmetric (31.13), RWCE (31.08) > CE (30.45) > PGL (30.36) > Sharpe (29.27). Paired t -tests confirm significance (MADL $p = 0.006$, $d = 1.09$; Asymmetric $p = 0.004$, $d = 1.02$; RWCE $p = 0.011$, $d = 1.03$; Sharpe $p < 0.001$, $d = -1.95$). Effect sizes are smaller than on BTC ($d \approx 1.0$ vs. $d > 2.0$): the MADL–CE gap halves (+0.77 vs. +1.46) and the MADL cross-seed std rises from 0.30 (BTC) to 0.78 (ETH), accounting for the $\sim 2.4\times$ d -compression; the ranking and family-vs.-non-family ordering survive intact, and accuracy is preserved ($\sim 72\%$).

5.9. Walk-Forward Cross-Validation

We conduct 3-fold walk-forward cross-validation with expanding training windows: Fold 1 (Bear, test 2022-07 to 2023-03), Fold 2 (Recovery, test 2023-04 to 2023-12), Fold 3 (Bull, test 2024-04 to 2024-12). Six losses $\times$ 3 seeds $\times$ 3 folds = 54 runs.

Table 15: Walk-forward cross-validation results (mean over 3 seeds per fold). Each fold uses an expanding training window with a different market condition as test period. Sharpe ratios are annualized from hourly returns ($\times\sqrt{8760}$). Bold indicates best Sharpe per fold and overall.

	Bear (2022-07–2023-03)		Recovery (2023-04–2023-12)		Bull (2024-04–2024-12)		Mean	
Loss Function	Acc (%)	Sharpe	Acc (%)	Sharpe	Acc (%)	Sharpe	Acc (%)	Sharpe
CE	71.7	22.9	72.2	17.0	72.5	27.7	72.2	22.5
MADL	71.1	24.0	72.6	20.9	72.6	28.2	72.1	24.3
RWCE	71.1	24.0	72.3	20.4	72.6	28.2	72.0	24.2
Asymmetric	71.0	23.7	72.7	20.6	72.6	28.4	72.1	24.2
Profit-Guided	71.9	23.1	72.4	19.1	72.5	27.4	72.3	23.2
Sharpe Loss	71.0	23.8	70.9	18.2	71.7	27.1	71.2	23.0

393 All three return-weighted losses outperform CE in every fold (Sharpe
394 range 17–28). Paired t -tests on fold means do not reach significance ($n = 3$;
395 smallest $p = 0.18$), but Page’s trend test [28] on the predicted ordering
396 yields $L = 261$ of $L_{\max} = 273$, exact $p = 5.6 \times 10^{-4}$ on both Sharpe and ROI,
397 strongly rejecting random ranking.

398 **Per-fold τ sensitivity (extended grid).** Re-running MADL over the
399 extended grid $\tau \in \{1, 2, 3, 5, 7, 10, 15\}$ on each fold (3 seeds, 63 runs; Ta-
400 ble 16): within-fold peak-to-peak Sharpe spread across $\tau \in [2, 15]$ is 1.10
401 (Bear), 1.14 (Recovery), 0.76 (Bull), each within cross-seed std (0.3–1.8); the
402 $\operatorname{argmax} \tau^*$ ($\tau = 2, 5, 15$) is uncorrelated with fold κ (17.5, 91.8, 13.5). Only
403 $\tau = 1$ is consistently weakest, 0.3–1.2 Sharpe points below the plateau. The
404 practical conclusion: τ is a low-stakes hyperparameter anywhere in $[2, 15]$.

405 **Alternative weighting schemes (default HP).** GMADL [29] ($w(r, \hat{r}) =$
406 $-(\sigma(ar\hat{r}) - 0.5)|r|^b$, defaults $a = 10, b = 2$) and Confidence-Weighted LRW
407 (CL-RW, $\mathcal{L}_{\text{RW}}$ weights $\times(2p_{\max} - 1)$) at 3 seeds (Table G.23). CL-RW ties

Table 16: Extended per-fold τ -sweep (MADL, 3 seeds; κ is the test-period excess kurtosis). Bold marks the per-fold argmax; all $\tau \geq 2$ are statistically indistinguishable within each fold.

Fold (κ)	$\tau = 1$	$\tau = 2$	$\tau = 3$	$\tau = 5$	$\tau = 7$	$\tau = 10$	$\tau = 15$
Bear ($\kappa = 17.5$)	23.66 ± 0.44	24.41 ± 0.47	24.00 ± 0.56	24.35 ± 0.32	23.61 ± 0.46	23.78 ± 0.84	23.31 ± 0.69
Recovery ($\kappa = 91.8$)	19.25 ± 1.03	20.42 ± 1.28	19.87 ± 1.12	20.68 ± 1.76	20.31 ± 1.06	19.54 ± 1.39	19.95 ± 1.62
Bull ($\kappa = 13.5$)	27.27 ± 1.62	28.48 ± 0.38	28.30 ± 0.30	28.06 ± 1.29	28.08 ± 0.73	28.69 ± 0.31	28.82 ± 0.50

MADL ($\Delta\text{Sharpe} = -0.66$, $p = 0.103$); default-HP GMADL falls to accuracy
 $\sim 62\%$ ($\Delta\text{Sharpe} = -6.27$, $p = 0.010$). GMADL’s defaults were calibrated to
lower-frequency equities; recovery under a 2D (a, b) search is plausible and is
left as future work—this is therefore a default-HP transfer-failure result, not
a fair head-to-head (Appendix Appendix G).

Mechanism: empirical noise- κ coupling. A bias-variance account
predicts $\tau^* = \arg \min_{\tau} [(1 + \lambda)\mathbb{E}[w^2] - 2\mathbb{E}[wu]]$, with τ^* rising monotonically
in κ at fixed λ on synthetic Student- t data (Pearson $r \in [0.93, 0.98]$;
src/tau_theory_synthetic.py). Empirically flat τ^* is therefore consistent
with theory only if real-market λ rises with κ . Using cross-seed Sharpe
std per (loss, fold) cell as a finite-sample λ proxy: across $n = 18$ cells,
 $\text{corr}(\log \kappa, \text{std}_{\text{seed}} \text{Sharpe}) = +0.633$, $p = 0.005$ (Spearman $+0.603$, $p =$
 0.008), while classification error is flat in κ ($r = -0.126$, $p = 0.62$; Ap-
pendix Appendix I). Fold-mean re-aggregation (independent $n = 3$) yields
 $r = +0.9995$, $p = 0.020$; the within-loss Spearman is positive at all six losses.
Accuracy is regime-invariant but downstream Sharpe variance scales with
tail heaviness—the coupling the theory requires. τ -robustness is therefore
the joint consequence of (i) clip+renormalize making τ^* a function of κ/λ
and (ii) real markets supplying a κ - λ coupling that holds the ratio approxi-

427 mately constant.

428 6. Decision Support Framework for Loss Function Selection

429 Algorithm 1 maps practitioner inputs (objective, risk tolerance, trans-
430 action cost, directional preferences) to a loss and hyperparameters. Each
431 branch derives from a specific table: the $c > 0.005$ warning threshold from
432 Table 12; $\tau = 3$ for CONSERVATIVE from Table 10; asymmetric λ defaults
433 from ablation configurations that retained positive Sharpe.

434 6.1. Selection Algorithm

435 6.2. Recommendation Rationale

436 Table 17 summarizes the empirical evidence supporting each recommen-
437 dation.

438 6.3. Practical Guidelines and Integration

439 Return-weighted losses should replace CE as the default given negligible
440 implementation cost. Sharpe Loss is recoverable under a two-dimensional
441 grid search (β , warmup; Appendix Appendix C) but more expensive to tune
442 than RW's single τ . Algorithm 1 is deterministic and side-effect-free; List-
443 ing 1 is a reference Python implementation. Figure 8 situates the selector
444 within an end-to-end pipeline.

445 **Scope.** Branch rules are calibrated on BTC perpetual and validated on
446 ETH and Binance BTC spot; transfer to lower- κ markets is plausible but
447 unvalidated. The $c > 0.005$ warning threshold is the empirical break-even
448 in Table 12, not a tuning parameter. The hourly-annualized Sharpe used by
449 Algorithm 1 preserves the MADL/RWCE ranking under daily aggregation

Table 17: Decision framework for loss function selection. Recommendations are based on 10-seed experiments over 5+ years of BTC/USD data and validated on ETH/USDT (Section 5.8).

Trading tive	Objec-	Recommended Loss	Rationale
General-purpose (default)		RWCE ($\tau=3$)	Top-tier Sharpe (28.52), highest accuracy (72.31%), second-lowest std (0.38 vs. MADL 0.30, Asymmetric 0.69), single HP
Max Sharpe (ag- gressive)		MADL ($\tau=5$)	Top-tier Sharpe (28.53); tiebreakers: lowest std (0.30), shallowest drawdown (-2.34%), strongest down-period advantage ($+5.7\%$)
Max Sharpe (con- servative)		RWCE ($\tau=3$)	Top-tier Sharpe tied with MADL/Asymmetric, second-lowest std (0.38), highest accuracy
Asymmetric preference	risk	Asymmetric Directional	Configurable $\lambda_{up}/\lambda_{down}$; Sharpe 28.56 (top-tier tied)
Maximum accuracy only		Cross-Entropy	No financial HPs; suitable when post-hoc risk management handles profitability

Algorithm 1 Loss Function Selection for Cryptocurrency Trading Systems

Require: objective $\mathcal{O} \in \{\text{GENERAL}, \text{MAXSHARPE}, \text{ASYMMETRIC}, \text{MAXACCURACY}\};$

risk $\mathcal{R} \in \{\text{CONSERVATIVE}, \text{AGGRESSIVE}\}$ (default CONS.); cost c ; directional

$\lambda_{\text{up}}, \lambda_{\text{down}} \geq 1.0$

Ensure: Loss $\mathcal{L}$ with hyperparameters θ

```
1: if  $c > 0.005$  then
2:   warn: unprofitable above 0.5% per-trade cost (Table 12)
3: end if
4: if  $\mathcal{O} = \text{GENERAL}$  then
5:   return ReturnWeightedCE,  $\{\tau = 3\}$     ▷ Best accuracy–Sharpe–variance
      trade-off
6: else if  $\mathcal{O} = \text{MAXSHARPE} \wedge \mathcal{R} = \text{AGGRESSIVE}$  then
7:   return MADL,  $\{\tau = 5\}$     ▷ Top-tier Sharpe, shallowest drawdown
8: else if  $\mathcal{O} = \text{MAXSHARPE}$  then
9:   return ReturnWeightedCE,  $\{\tau = 3\}$     ▷ Top-tier Sharpe, lowest cross-seed
      std in trio
10: else if  $\mathcal{O} = \text{ASYMMETRIC}$  then
11:   if downside risk aversion then
12:     return AsymDir,  $\{\lambda_{\text{down}}=2.0, \lambda_{\text{up}}=1.0, \tau_{\text{mag}}=3.0\}$ 
13:   else
14:     return AsymDir,  $\{\lambda_{\text{up}}=2.0, \lambda_{\text{down}}=1.0, \tau_{\text{mag}}=3.0\}$ 
15:   end if
16: else if  $\mathcal{O} = \text{MAXACCURACY}$  then
17:   return CrossEntropy,  $\emptyset$ 
18: end if
```

(Table 5), but Asymmetric flips sign (+5.5% hourly vs. −1.7% daily, 3 seeds), so we recommend evaluating both metrics before deployment. For relative loss ranking, per-bar bp is unambiguous (Section 4.7).

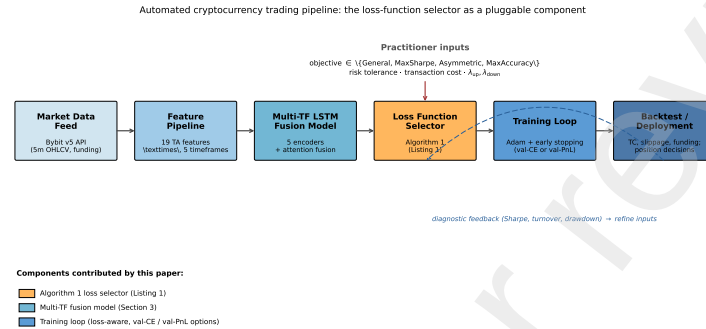


Figure 8: Pipeline view: Algorithm 1 (orange) plugs into market-data, feature, fusion, and backtest stages.

Listing 1: Reference Python implementation of Algorithm 1; returns (loss_class, hyperparams) for the training loop.

```

453
454 1 import warnings
455 2
456 3 def select_loss(objective, risk="Conservative", cost
457     =0.001,
458     lambda_up=1.0, lambda_down=1.0):
459 4     if cost > 0.005:
460 5         warnings.warn("All strategies unprofitable
461             above 0.5% per-trade cost")
462 6     if objective == "General":
463 7         return "ReturnWeightedCE", {"max_weight": 3.0}
464 8     if objective == "MaxSharpe":
465 9         return ("MADL", {"max_weight": 5.0}) if risk

```

```

466         == "Aggressive" \
467 11         else ("ReturnWeightedCE", {"max_weight":
468         3.0})
469 12     if objective == "Asymmetric":
470 13         return "AsymmetricDirectional", {"lambda_up":
471         lambda_up,
472 14         "lambda_down": lambda_down, "max_weight":
473         3.0}
474 15     if objective == "MaxAccuracy":
475 16         return "CrossEntropy", {}
476 17     raise ValueError(f"Unknown objective: {objective}")
477     )
478

```

479 7. Discussion

480 7.1. Why Return-Weighting Works

481 Three mechanisms: (1) implicit curriculum learning [30]—up-weighting
482 large moves prioritises high-signal samples; (2) noise reduction—down-weighting
483 small moves attenuates microstructure noise; (3) alignment with the evalu-
484 ation metric—trading PnL is inherently return-weighted. The empirically
485 wide $\tau \in [2, 15]$ plateau reflects the κ - λ coupling: the bias-variance trade-
486 off between PnL alignment (larger τ) and gradient concentration (smaller τ)
487 cancels when tail kurtosis and gradient noise rise together.

488 7.2. Why Sharpe Loss and Profit-Guided Loss Underperform

489 **Sharpe Loss:** gradient conflict with CE; 1.44pp accuracy drop ($p <$
490 0.001). Aggressive tuning ($\beta = 0.3$, $w = 0$, Appendix Appendix C) recovers

491 RW-level performance but requires a 2D HP search. **PGL**: under $w = 5$,
 492 all four configs select epoch 6 with effective $\alpha = 0$ (Appendix Appendix B);
 493 the selected model is essentially pure CE. Even with $w = 0$, val-CE-driven
 494 selection locks in a CE-dominated solution; val-PnL-driven selection lifts
 495 Sharpe only from 27.04 to 27.24, still below MADL/RWCE. The selection
 496 criterion must align with the auxiliary objective.

497 7.3. Statistical vs. Economic Significance

498 Large Cohen’s d (> 2.0) reflects low between-seed variance rather than
 499 large absolute gains: per-bar net return improves by 0.72 bp (+4.9%) and
 500 Sharpe by ~ 1.5 points. The 58% cross-seed variance reduction (MADL vs.
 501 CE) is arguably more valuable than the mean improvement for position siz-
 502 ing.

503 7.4. Limitations

- 504 1. **Cross-asset breadth.** BTC + ETH only; SOL/XRP and a low- κ
 505 control (e.g. S&P 500 ETF) are needed to externalise the κ - λ coupling.
- 506 2. **Fixed architecture.** Transformer/gated-fusion backbones are future
 507 work.
- 508 3. **Feature normalization.** Global StandardScaler; rolling normaliza-
 509 tion unexplored.
- 510 4. **Cost model.** Fixed slippage 0.05% (volatility-proportional in Sec-
 511 tion 5.4); funding payments omitted ($\pm 7.36\%$ cumulative, ranking pre-
 512 served).
- 513 5. **Limited HP optimization.** PGL early-stopping patience > 10 not
 514 tested.

- 515 **6. Limited temporal validation.** 3-fold $\times$ 3 seeds; more folds would
516 enable a properly-identified mixed-effects test.
- 517 **7. Single-seed leakage diagnostic.** Signal-decay (Figure 1) on one
518 seed.
- 519 **8. ETH period confound.** The smaller $d \approx 1$ decomposes into a halved
520 gap (+0.77 vs. +1.46) and a $2.6\times$ larger cross-seed std (0.78 vs. 0.30).
- 521 **9. Position sizing.** Full-size positions throughout; Kelly/vol-targeting
522 is a concrete next step.
- 523 **10. Comparators not covered.** No direct comparison against differen-
524 tiable Sharpe approximations beyond Moody and Saffell [9], distributional-
525 regression, or focal losses.

526 **8. Conclusion**

527 Over 5 years of BTC/USD with 10 seeds per loss, three findings emerge.
528 **First**, conditional on multi-timeframe fusion supplying $\sim 72\%$ baseline ac-
529 curacy, return-magnitude-weighted losses (MADL, RWCE, Asymmetric) im-
530 prove per-bar net return from 14.83 to 15.55–15.57 bp (+4.9–5.0%; $p <$
531 0.001 Bonferroni-corrected) and hourly-annualized Sharpe from 27.07 to 28.5
532 (+5.4–5.5%); the daily-aggregated Sharpe gain is +0.4–3.1% across the trio.
533 The ranking survives Newey–West HAC correction (borderline $p = 0.054$
534 for MADL). **Second**, more aggressive financial optimization (Sharpe Loss,
535 PGL) underperforms at literature defaults; Sharpe Loss recovers under 2D
536 grid search, PGL does not. **Third**, return-weighted losses reduce cross-
537 seed per-bar-bp variance by 58% (MADL vs. CE), an implicit regulariza-
538 tion effect. We recommend RWCE as the default, with τ tuning low-stakes

539 within $[2, 15]$ given the κ - λ coupling. Future work: data-driven τ rules,
540 additional altcoins/lower- κ controls, Kelly/vol-targeting deployment simula-
541 tion, live trading, and comparison against differentiable-Sharpe and focal-loss
542 variants.

543 **Appendix A. PGL Hyperparameter Search**

544 Grid: $\alpha \in \{0.1, 0.3, 0.5, 1.0, 2.0\}$, warmup $w \in \{0, 3, 5, 10\}$ (20 configs $\times$
545 3 seeds). Best ($\alpha = 1.0$, $w = 0$): Sharpe = 27.0, far below MADL/RWCE
546 (28.5). $w = 10$ produces identical results across α (selected model from
547 pure-CE warmup). PGL's underperformance is fundamental.

Table A.18: Profit-Guided Loss hyperparameter grid search (mean over 3 seeds). The default configuration ($\alpha = 0.5$, warmup = 5) is marked with †. The best configuration is shown in bold. The CE baseline over the same 3 seeds achieves Sharpe = 26.42; the best PGL configuration ($\alpha = 1.0$, warmup = 0) marginally exceeds it (Sharpe = 27.0, +2.2%), but this improvement is within the CE 10-seed standard deviation (± 0.72). Configurations with warmup = 10 yield identical results across all α values because early stopping selects the model from the pure-CE warmup phase.

α	Warmup	Accuracy (%)	Sharpe	ROI (%)	MaxDD (%)
0.1	0	72.27	26.6	3364	-2.34
0.1	3	72.22	26.5	3352	-2.35
0.1	5	72.07	26.4	3347	-2.36
0.3	0	72.27	26.6	3365	-2.25
0.3	3	72.23	26.9	3393	-2.32
0.3	5	72.07	26.4	3347	-2.36
0.5	0	72.29	26.4	3344	-2.29
0.5	3	72.27	26.6	3362	-2.38
0.5†	5	72.13	26.6	3362	-2.34
1.0	0	72.32	27.0	3413	-2.37
1.0	3	72.19	26.5	3351	-2.36
1.0	5	72.15	26.6	3360	-2.34
2.0	0	72.32	26.9	3394	-2.41
2.0	3	72.24	27.0	3405	-2.32
2.0	5	72.15	26.6	3360	-2.16
<i>warmup = 10 (all α identical):</i>					
any	10	70.89	24.4	3105	-2.69

548 Appendix B. PGL Learning Curves

549 Four configurations (CE, PGL $\alpha=0.5/w=5$, $\alpha=0.5/w=0$, $\alpha=1.0/w=0$)
 550 all select best epoch = 6; under $w = 5$ this coincides with effective $\alpha = 0$.

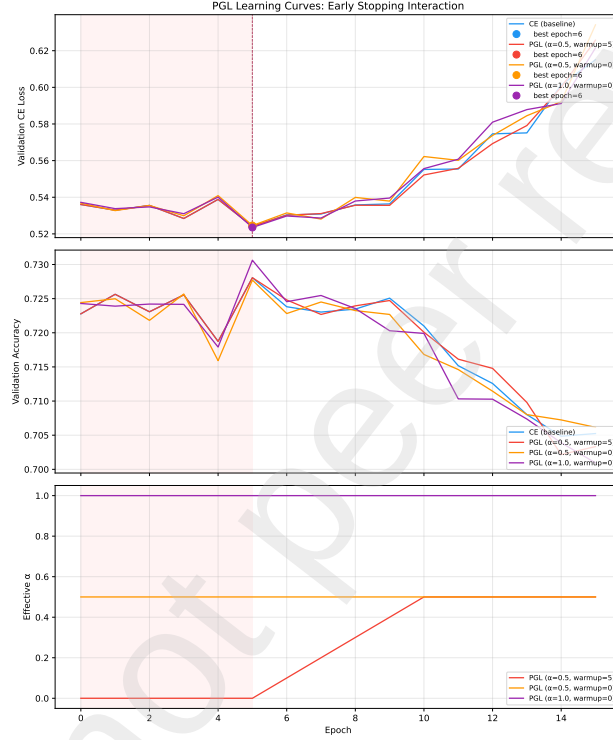


Figure B.9: PGL learning curves. All four select epoch 6; under $w = 5$, $\alpha = 0$ at best epoch.

551 Appendix C. Sharpe Loss Hyperparameter Search

552 Grid: $\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, $w \in \{0, 3, 5, 10\}$. Best ($\beta = 0.3$, $w = 0$):
 553 Sharpe = 28.00 (+6.0% vs. 3-seed CE baseline, approaching MADL/RWCE
 554 28.5). HP-recoverable but requires 2D search.

Table C.19: Sharpe Loss hyperparameter search (mean $\pm$ std over 3 seeds). CE baseline: Sharpe = 26.42, Accuracy = 0.7220.

CE Floor (β)	Warmup	Accuracy	Sharpe (annual)
0.1	0	0.7123 $\pm$ 0.0010	27.80 $\pm$ 0.40
0.1	3	0.7158 $\pm$ 0.0029	27.45 $\pm$ 0.62
0.1	5	0.7104 $\pm$ 0.0015	26.98 $\pm$ 0.45
0.1	10	0.6937 $\pm$ 0.0042	23.65 $\pm$ 0.21
0.3	0	0.7170 $\pm$ 0.0036	28.00 $\pm$ 0.60
0.3	3	0.7188 $\pm$ 0.0016	26.99 $\pm$ 0.43
0.3	5	0.7086 $\pm$ 0.0026	26.14 $\pm$ 0.81
0.3	10	0.6875 $\pm$ 0.0019	22.50 $\pm$ 0.23
0.5	0	0.7158 $\pm$ 0.0051	27.92 $\pm$ 0.63
0.5	3	0.7174 $\pm$ 0.0022	26.87 $\pm$ 0.61
0.5	5	0.7090 $\pm$ 0.0050	26.10 $\pm$ 1.25
0.5	10	0.6870 $\pm$ 0.0009	21.59 $\pm$ 0.92
0.7	0	0.7230 $\pm$ 0.0021	26.95 $\pm$ 0.29
0.7	3	0.7146 $\pm$ 0.0036	26.18 $\pm$ 0.83
0.7	5	0.7093 $\pm$ 0.0016	25.03 $\pm$ 0.52
0.7	10	0.6864 $\pm$ 0.0041	20.87 $\pm$ 0.11
0.9	0	0.7217 $\pm$ 0.0045	26.31 $\pm$ 0.33
0.9	3	0.7155 $\pm$ 0.0053	27.04 $\pm$ 0.44
0.9	5	0.7097 $\pm$ 0.0013	24.59 $\pm$ 0.62
0.9	10	0.7048 $\pm$ 0.0080	24.62 $\pm$ 1.20

555 Appendix D. Shuffle Diagnostic

556 Seeds 42–44, $N_{\text{test}} = 23,040$. All shuffled-label predictions assign to “up”;
 557 seed-44 std 0.005 is FP32 noise.

Table D.20: Shuffle diagnostic. Shuffled-label model collapses to $P(\hat{y} = \text{up}) = 1$ with near-constant softmax (std ≈ 0.002); 50.79% is exactly the up-class proportion.

Regime	Seed	Acc	$P(\hat{y} = \text{up})$	Softmax std	Confusion (TN/FP/FN/TP)
Normal	42	0.7229	0.461	0.268	8688/2649/3736/7967
Normal	43	0.7255	0.516	0.260	8076/3261/3064/8639
Normal	44	0.7178	0.521	0.273	7934/3403/3099/8604
Shuffled	42	0.5079	1.000	0.000	0/11337/0/11703
Shuffled	43	0.5079	1.000	0.000	0/11337/0/11703
Shuffled	44	0.5079	1.000	0.005	0/11337/0/11703

558 Appendix E. Bootstrap Cohen’s d Confidence Intervals

559 Bootstrap 95% CIs for paired Cohen’s d on Sharpe (vs. CE; 10,000 resam-
 560 ples over 10 seeds). MADL/RWCE/Asymmetric all exceed the large-effect
 561 threshold ($d > 0.8$) by 1.6–2.5 $\times$.

562 Appendix F. Friedman Test (Supplementary)

563 Friedman–Nemenyi [24] assumes independent ranks across datasets; our
 564 repeated-runs setup underestimates uncertainty. Reported for completeness
 565 only.

Table E.21: Bootstrap 95% CIs for paired Cohen's d on Sharpe (vs. CE).

Loss	Cohen's d	95% CI (lower)	95% CI (upper)
MADL	2.63	1.97	5.04
RWCE	2.51	1.78	4.40
AsymmetricDirectional	2.09	1.26	4.08

Table F.22: Friedman ranks across 10 seeds; CD is Nemenyi critical difference at $\alpha = 0.05$.

Metric	χ^2	p -value	CD	Best (rank)
Accuracy	25.43	1.15×10^{-4}	2.38	RWCE (2.5)
F1	25.23	1.26×10^{-4}	2.38	RWCE (2.4)
Sharpe	41.31	8.12×10^{-8}	2.38	RWCE (1.9)
ROI	41.31	8.12×10^{-8}	2.38	RWCE (1.9)

566 Appendix G. Alternative Weighting Schemes (Pilot)

Table G.23: GMADL ($a=10$, $b=2$) and Confidence-Weighted LRW vs. MADL on the main split (3 seeds; paired t -test). CL-RW multiplies the $\mathcal{L}_{\text{RW}}(\tau=5)$ weight by $(2p_{\text{max}} - 1)$. GMADL collapses to near-prior accuracy because $|r| \sim 10^{-2}$ keeps $ar\hat{r}$ in the near-linear region of the sigmoid and $|r|^b \sim 10^{-4}$ flattens the landscape; recovery would require a separate grid search over a, b .

Loss	Acc	Sharpe	$\Delta\text{Sharpe vs MADL}$	p vs MADL
MADL	0.7212 ± 0.0029	28.42 ± 0.33	—	—
RWCE	0.7233 ± 0.0023	28.20 ± 0.37	-0.21	0.571
ConfidenceWeightedLRW	0.7172 ± 0.0037	27.75 ± 0.47	-0.66	0.103
CE	0.7220 ± 0.0039	26.42 ± 0.57	-1.99	0.014^*
GMADL ($a=10$, $b=2$)	0.6166 ± 0.0061	22.14 ± 1.42	-6.27	0.010^*

567 Appendix H. Cross-Seed Boxplot

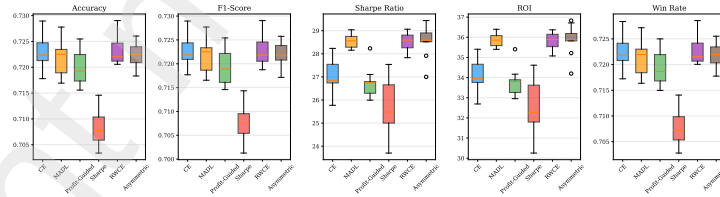


Figure H.10: 10-seed Sharpe distributions. Return-weighted variants show tighter distributions, consistent with implicit regularization.

568 Appendix I. Empirical Noise–Kurtosis Coupling

569 Each point is one (loss, fold) cell across the 3-fold walk-forward data
 570 ($n = 18$).

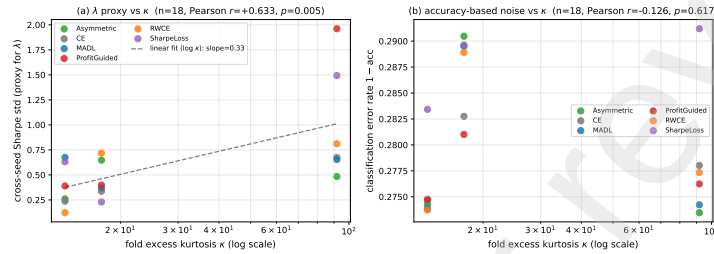


Figure I.11: (a) cross-seed Sharpe std (λ proxy) vs. fold κ : Pearson $r = +0.633$, $p = 0.005$.
 (b) classification error rate vs. κ : $r = -0.126$, $p = 0.62$.

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577 Declaration of Competing Interest

578 The author declares no known competing financial interests or personal
 579 relationships that could have appeared to influence the work reported in this
 580 paper.

581 **CRediT Author Statement**

582 **Haruki Tanabe:** Conceptualization, Methodology, Software, Validation,
583 Formal analysis, Investigation, Data Curation, Writing – Original Draft,
584 Writing – Review & Editing, Visualization.

585 **Data Availability**

586 BTC/ETH OHLCV data are obtainable from the public Bybit v5 (<https://bybit-exchange.github.io/docs/v5/intro>) and Binance Spot APIs;
587 endpoints and date ranges are documented in the experimental setup. Trained
588 model artefacts, numerical result CSVs, and the training/evaluation/analysis
589 code are available from the author on reasonable request.
590

591 **Declaration of Generative AI and AI-assisted technologies in the** 592 **writing process**

593 During the preparation of this work the author used a large language
594 model (Anthropic Claude Opus 4) to assist with manuscript editing, reviewer-
595 response anticipation, statistical sanity-checking, and LaTeX formatting. Af-
596 ter using this tool the author reviewed and edited the content as needed and
597 takes full responsibility for the content of the publication.

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